THE LANCET

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Hofman MS, Emmett L, Sandhu S, et al. [^{177}Lu]Lu-PSMA-617 versus cabazitaxel in patients with metastatic castration-resistant prostate cancer (TheraP): a randomised, open-label, phase 2 trial. *Lancet* 2021; published online Feb 11. http://dx.doi.org/10.1016/S0140-6736(21)00237-3.

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.
Supplement to: Hofman MS, Emmett L, Sandhu S, *et al.* ^{177}Lu-PSMA-617 versus cabazitaxel in patients with metastatic castration-resistant prostate cancer (TheraP): a randomised, open-label, phase 2 trial

TheraP Trial: Supplementary Appendix

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Section S1. Collaborators

Sites in which study was open to recruitment in Australia, number of men randomised (n), medical oncology and nuclear medicine principal investigators; other co-investigators and key contributors from site (in order of study opening date):

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Austin Health, Melbourne (n=14): A/Prof Andrew Weickhardt, Prof Andrew Scott / A/Prof Sze-Ting Lee; Aurora Poon, Kunthi Pathmaraj, John Sachinidis, Uwe Ackermann, Ben Nguyen and Kenneth Young

Monash Health, Melbourne (n=13): A/Prof Arun Azad / Dr Edmond Kwan, Dr Shakher Ramdave; Dr David Pook

Fiona Stanley Hospital (n=12): Dr Andrew Redfern, Dr Michael McCarthy / Dr William Macdonald; Elizabeth Bailey

Royal North Shore, Sydney (n=10): A/Prof Alex Guminski, Dr Edward Hsiao; Dr Geoff Schembri, A/Prof Paul Roach, Prof Dale Bailey

Liverpool Hospital, Sydney (n=8): Dr Wei Chua, Dr Peter Lin; Dr Adam Cooper, Dr Bavanthi Balakrishnar

Key sponsor and coordinating centre personnel:

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TheraP Executive Management Committee: Prof Michael Hofman, Prof Louise Emmett, Prof Ian Davis, Kate Ford, Ailsa Langford, A/Prof Andrew Martin, Margaret McJannett, Nisha Rana, Prof Martin Stockler, Dr Sonia Yip and Dr Alison Zhang,

Australasian Radiopharmaceutical Trials Network (ARTnet): Prof Dale Bailey, A/Prof Roslyn Francis, Prof Michael Hofman, Olga Kovacev, Dr Melissa Latter, Dr Doug Smyth

ANZUP Independent Data and Safety Monitoring Committee: Ian Olver, Peta Forder and David Goldstein.

PET Imaging Central Review: Jason Callahan, Amir Iravani, Michael S. Hofman, Peter MacCallum Cancer Centre; Louise Emmett, St Vincent's Hospital; special acknowledgement to Stephane Chauvie and Alex Stancu for support for WIDEN (Dixit Solution, Turin Italy)

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Figure S1. Trial schema

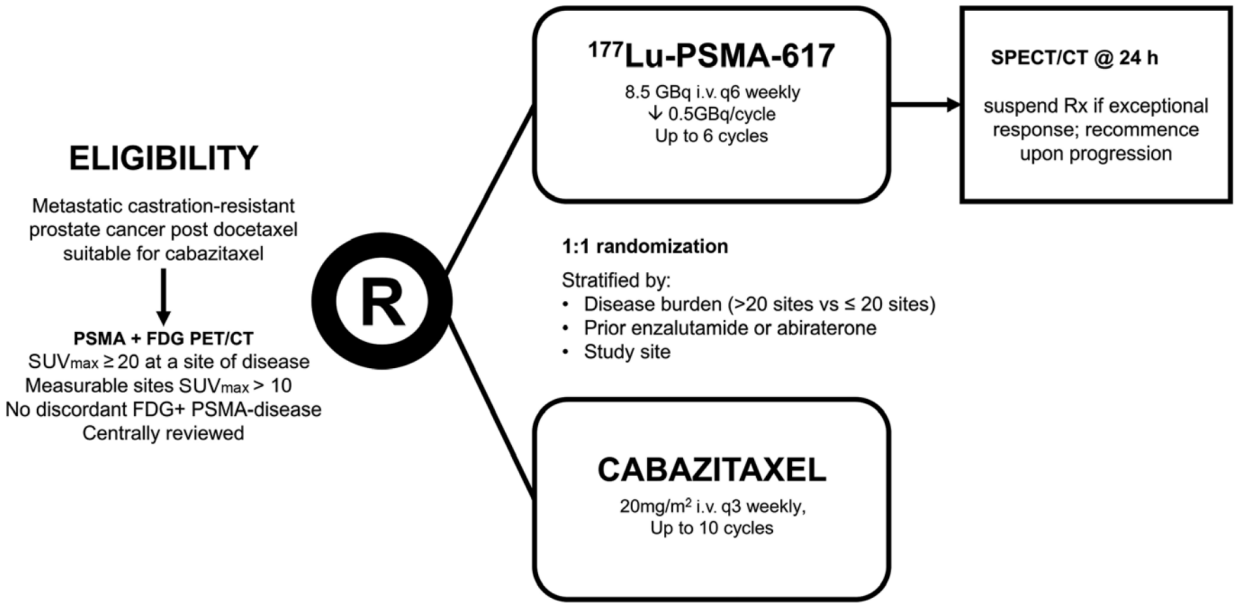
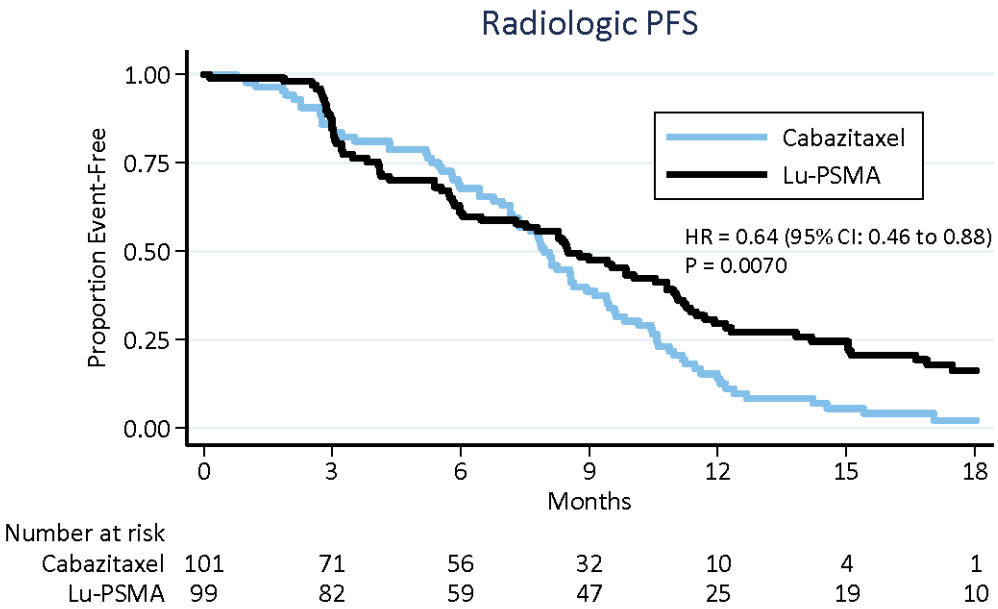


Figure S2. Radiologic progression free survival (rPFS) and PSA PFS



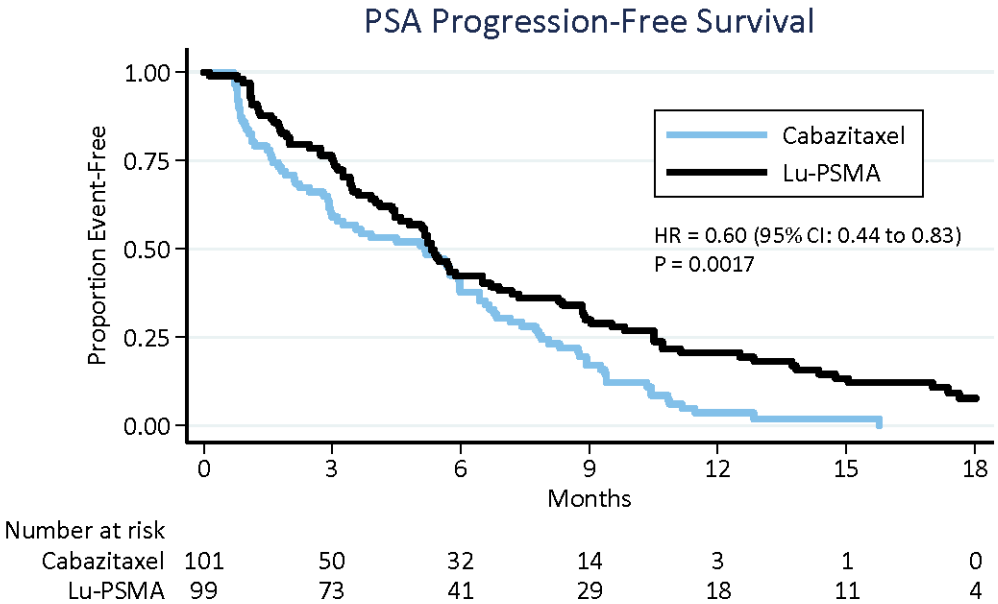


Figure S3. Change in Sum of Target Lesions for Patients with Baseline Measurable Disease according to RECIST 1.1

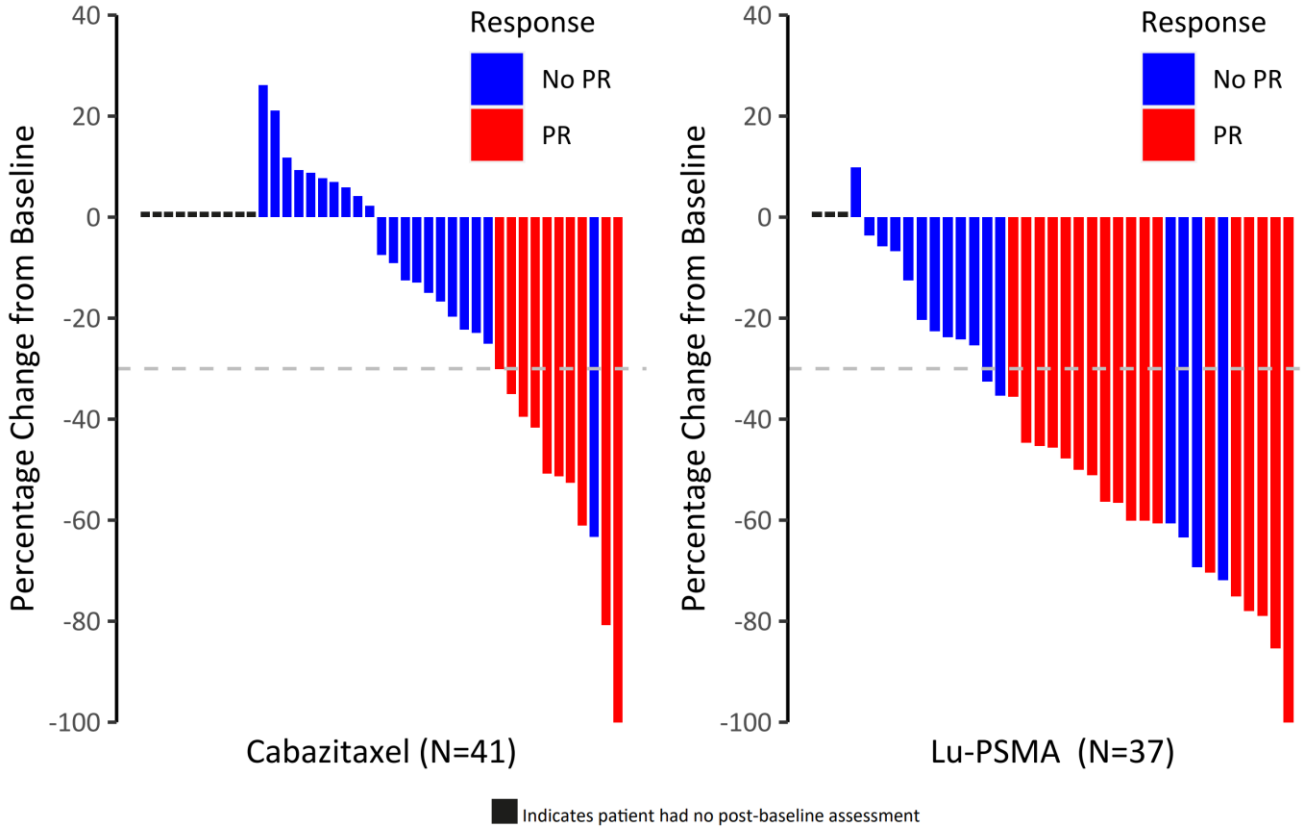
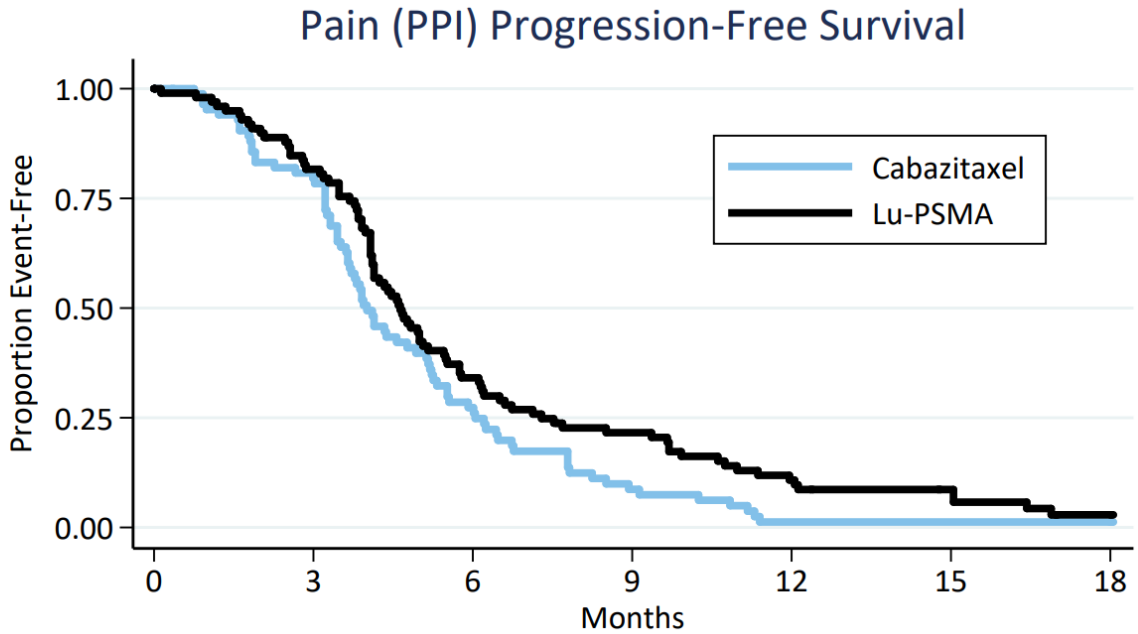


Figure S4. Pain (PPI) Progression-Free Survival*



Number at risk							
Cabazitaxel	85	66	22	7	1	1	1
Lu-PSMA	99	79	33	20	10	6	1

*HR 0·72; 95%CI 0·53-0·97; multi-variable Cox regression** P=0·0328; log-rank P=0·0553 (**study site and number of lesions)

Figure S5. EORTC QLQ-C30 patient reported outcomes by domains (1 of 3)

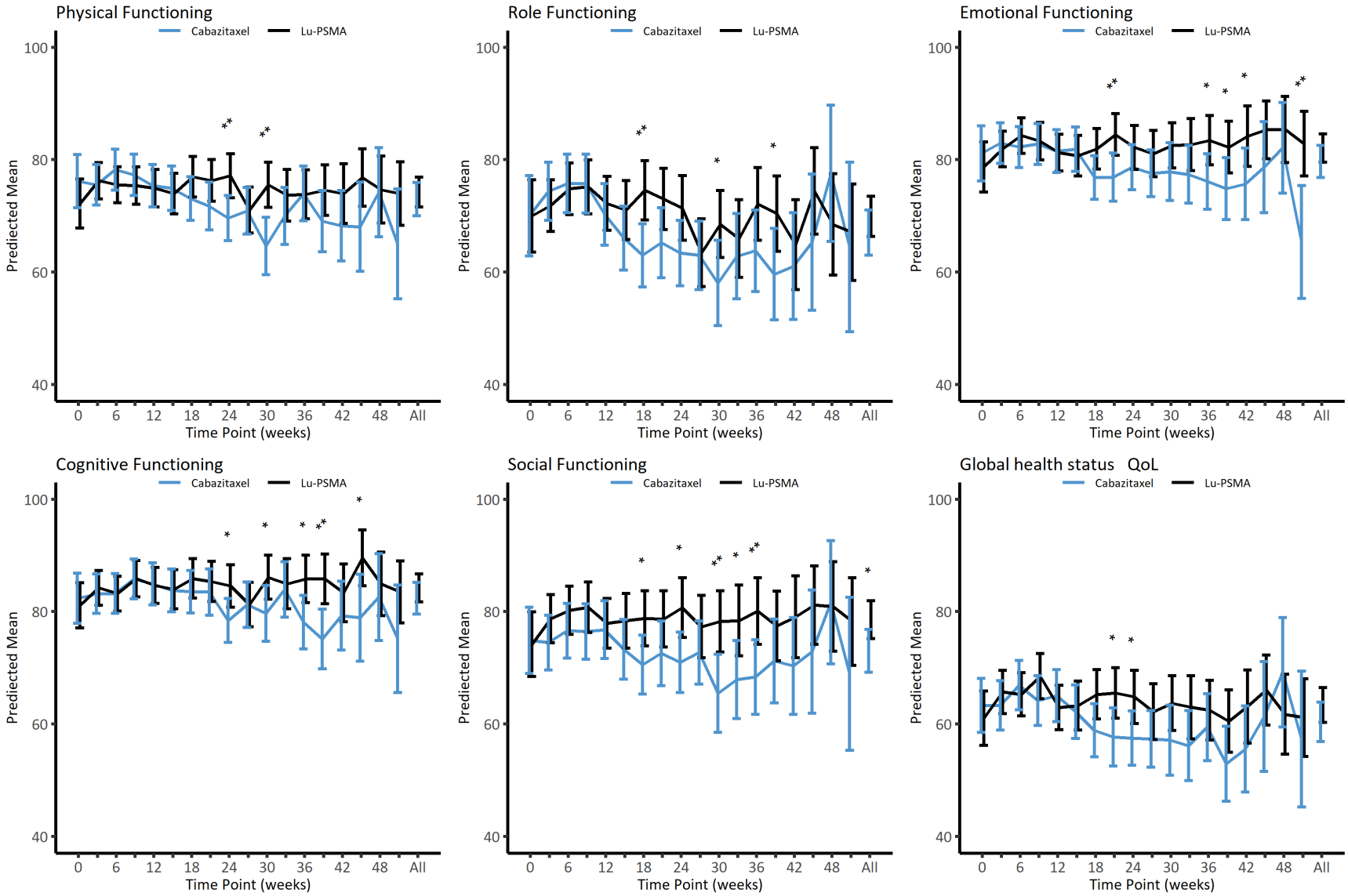


Figure S5. EORTC QLQ-C30 patient reported outcomes by domains (2 of 3)

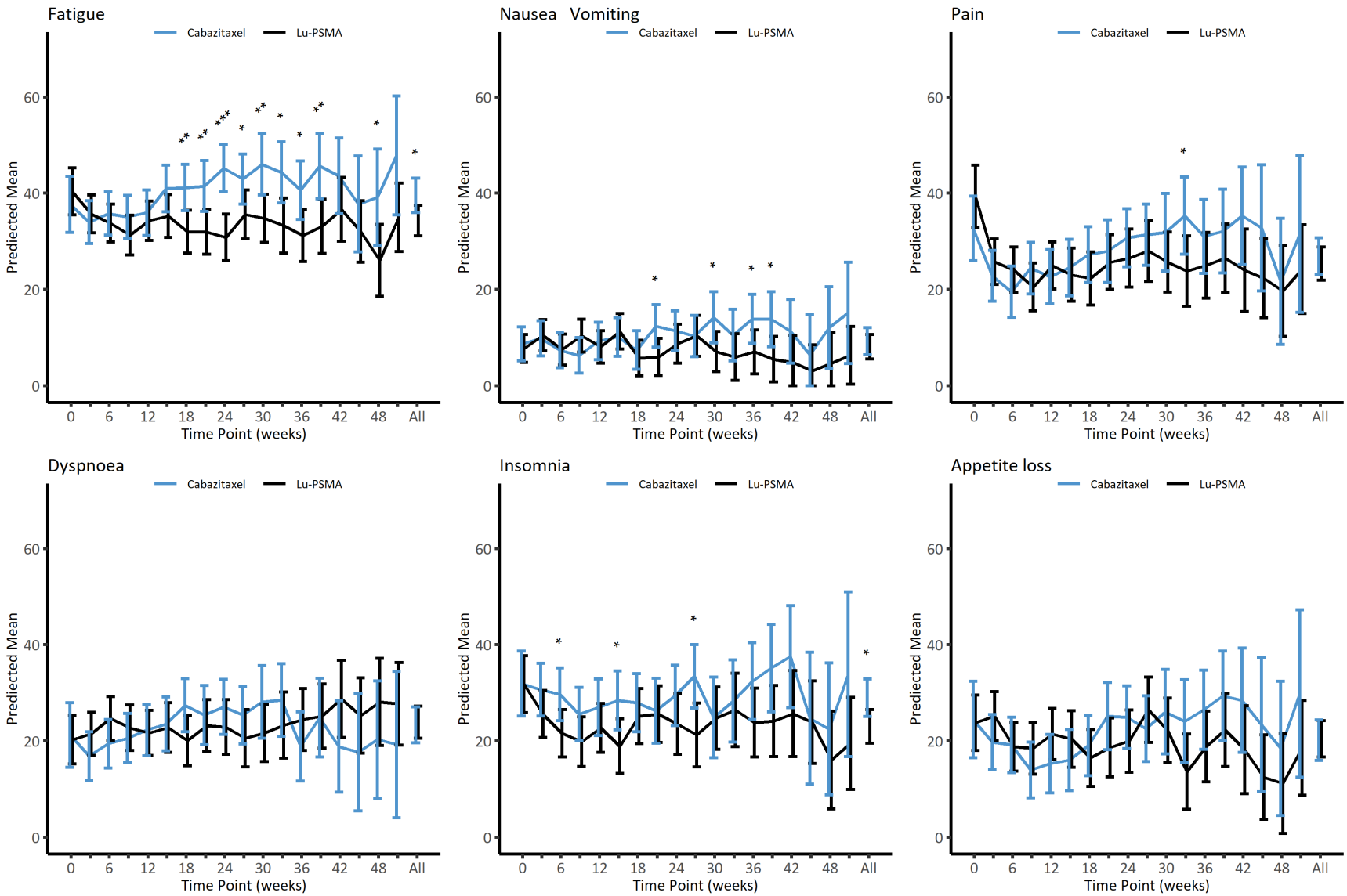
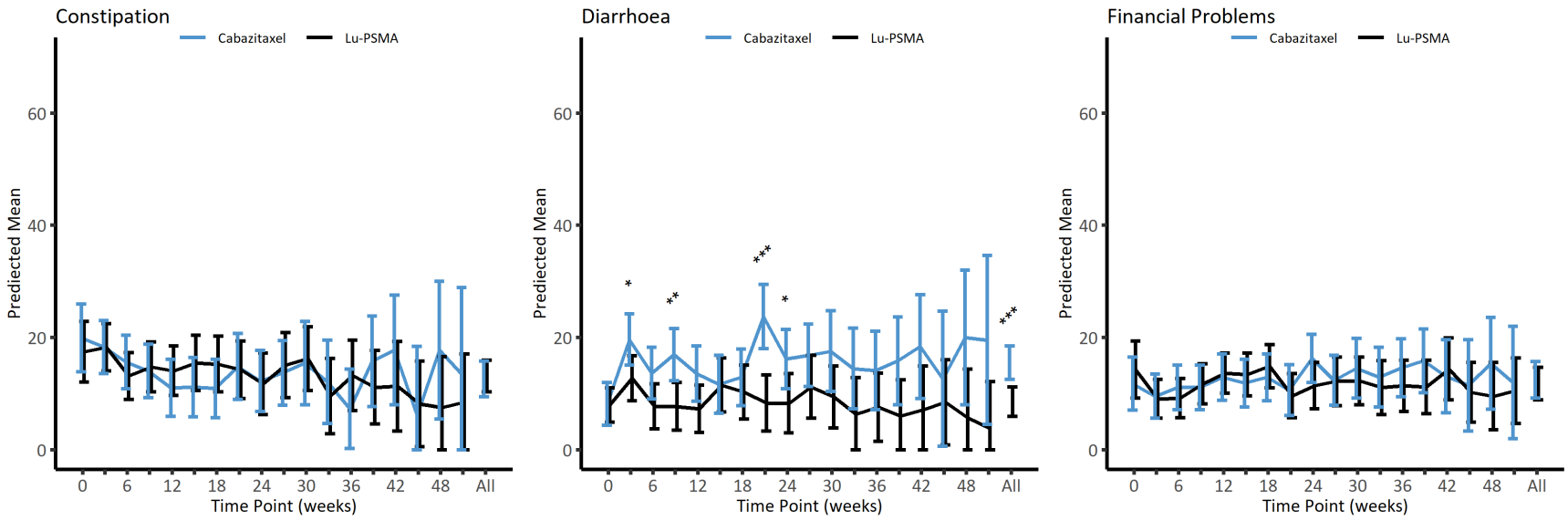


Figure S5. EORTC QLQ-C30 patient reported outcomes by domains (3 of 3)



*P < 0.05, ** P < 0.01, *** P < 0.001

Figure S6. Frequency of troublesome symptoms

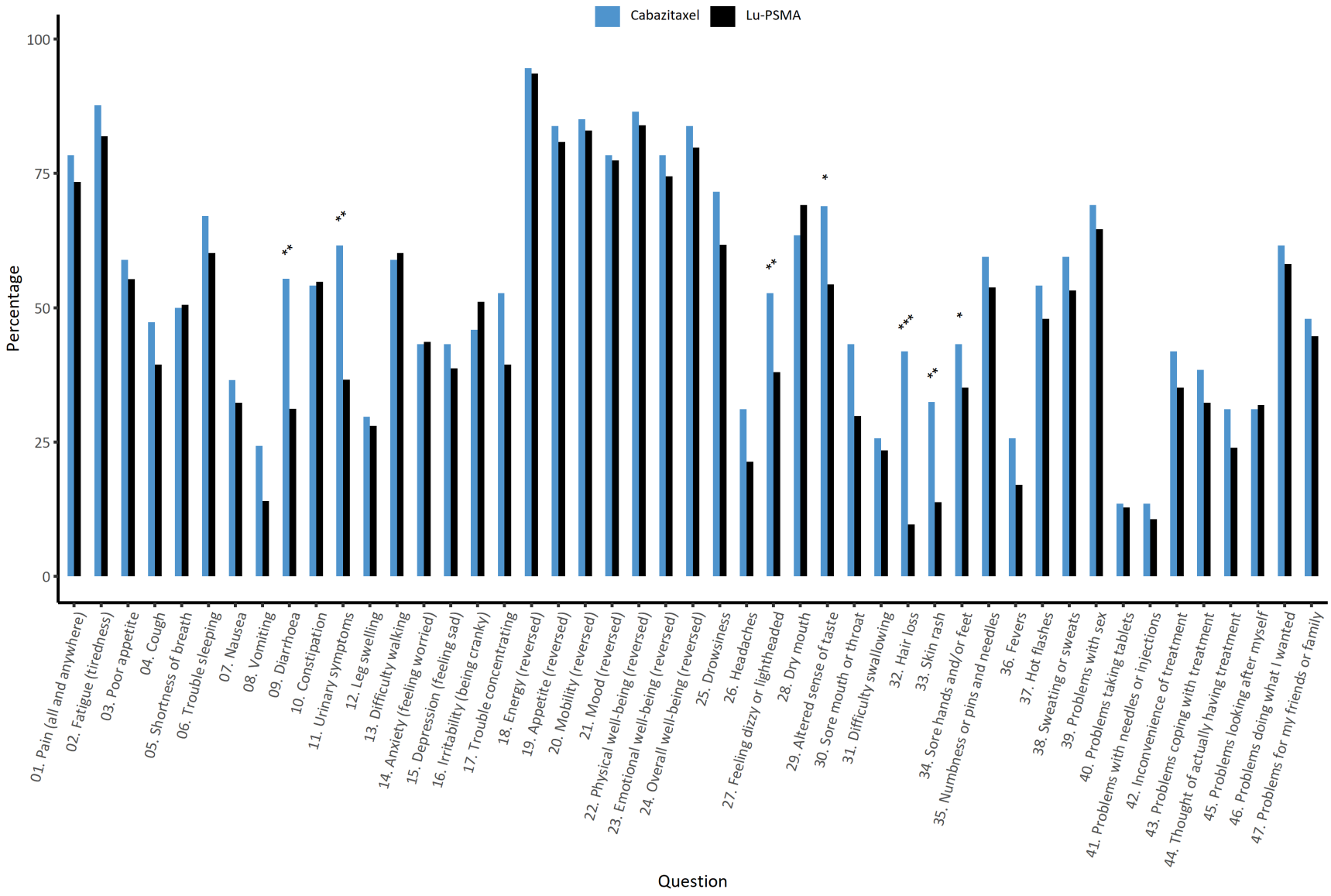
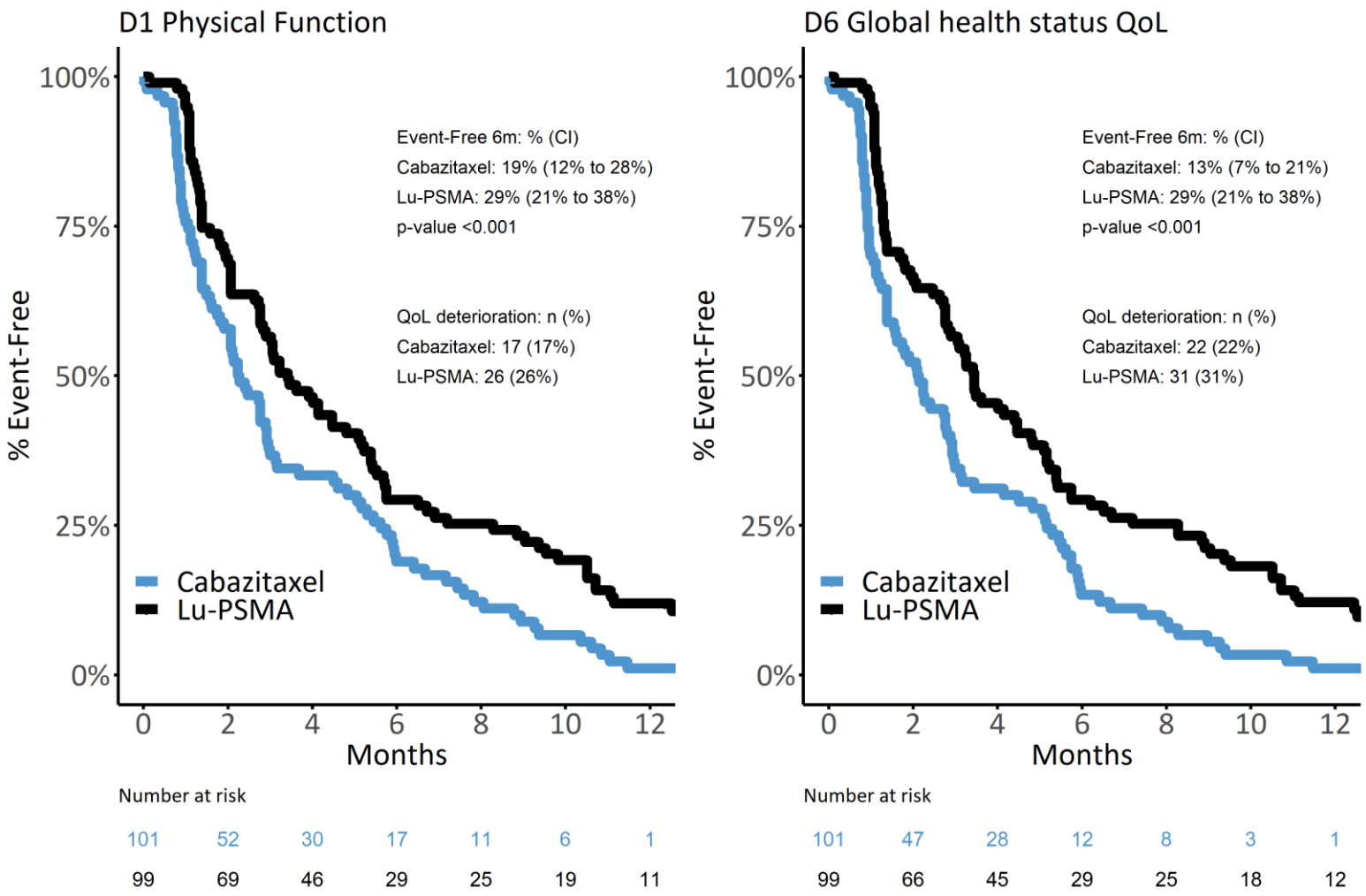


Figure S7. Kaplan-Meier Plot of Deterioration Free Survival*



*defined as the time from randomization to the first of the following events: a 10-point or more deterioration in health status from baseline (without subsequent 10-point or more improvement compared with baseline, or an improvement to ≥90 if the baseline score was ≥90), progression, death, or treatment discontinuation.

Table S1. Trial timeline

Date	Major event
31/10/2017	Protocol version 1.0
06/02/2018	Date first patient registered
30/07/2018	Amendment No. 1; enable follow-up of screen failed patients
03/09/2019	Last patient registered
31/12/2019	Data cut-off for first analysis (presented ASCO 2020)
20/07/2020	Data cut-off for Second analysis

Table S2. Eligibility criteria for enrolment

Inclusion criteria:

1. Male aged 18 or older with metastatic adenocarcinoma of the prostate defined by documented histopathology of prostate adenocarcinoma or metastatic disease typical of prostate cancer (i.e. involving bone or pelvic lymph nodes or para-aortic lymph nodes)
2. Castration-resistant prostate cancer, as defined as disease progressing despite castration by orchiectomy or ongoing Luteinizing Hormone-Releasing Hormone (LHRH) analog
3. Progressive disease with rising PSA defined by PCWG3 criteria (sequence of 2 rising values above a baseline at a minimum of 1-week intervals), and PSA ≥ 20 ng/mL
4. Target or non-target lesions according to RECIST 1.1
5. Prior treatment with docetaxel
6. Significant PSMA avidity on ^{68}Ga-PSMA PET/CT, defined as a minimum uptake of SUVmax 20 at a site of disease, and SUVmax > 10 at sites of measurable disease ≥10mm (unless subject to factors explaining a lower uptake, e.g. respiratory motion, reconstruction artefact)
7. ECOG Performance status 0 to 2
8. Assessed by a medical oncologist as suitable for chemotherapy with cabazitaxel
9. Adequate renal, bone and liver function
10. Estimated life expectancy > 12 weeks
11. Study treatment both planned and able to start within 21 days of randomisation
12. Willing and able to comply with all study requirements, including all treatments (cabazitaxel or Lu-PSMA); and, the timing and nature of all required assessments
13. Signed, written informed consent

Exclusion criteria:

1. Prostate cancer with known significant sarcomatoid or spindle cell or neuroendocrine small cell components
2. Site(s) of disease that are FDG positive with minimal PSMA expression defined as FDG intensity > ^{68}Ga-PSMA activity OR ^{68}Ga-PSMA SUVmax < 10

3. Sjogren's syndrome
4. Prior treatment with cabazitaxel or Lu-PSMA
5. Contraindications to the use of corticosteroid treatment
6. Active malignancy other than prostate cancer
7. Concurrent illness, including severe infection that may jeopardise the ability of the participant to undergo the procedures outlined in this protocol with reasonable safety
8. Serious psychological, familial, sociological or geographical condition that might hamper compliance with the study protocol and follow-up schedule
9. Patients who are sexually active and not willing/able to use medically acceptable forms of barrier contraception

Table S3. ^{68}Ga-PSMA-11 quality control specifications

Test	Specification
Validation testing QC*	
Visual inspection	Clear, colourless, particulate free
Product mass	≤ 30micrograms PSMA-11
pH	4-8
Radionuclide identity	T1/2 68 mins (62-74mins) 511keV energy peak
Radiochemical purity (TLC)	<3% colloidal form
Radiochemical purity (HPLC)	>91% 68Ga-PSMA-11 purity <2% 68Ga (III)ion
Ge-68 contamination	<0.001% 68Ge
Residual solvent analysis	Ethanol <100 000ppm Acetone <50000 ppm (if applicable)
Filter integrity	Meets filter manufacturers testing specification
Sterility test	Pass
Endotoxin test	<175 EU/dose
Batch release QC (for each synthesis)	
Visual inspection	Clear, colourless, particulate free
Label inspection	Product, volume, activity, date, time

Test	Specification
pH	4-8
Radionuclide identity	T1/2 68 mins (62-74mins) 511keV energy peak
Radiochemical purity (TLC)	<3% colloidal form
Radiochemical purity (HPLC)	>91% ^{68}Ga-PSMA-11 purity <2% ^{68}Ga (III)ion

*three consecutive successful validations required. PET camera site validation also undertaken by the Australasian Radiopharmaceutical Trials network (ARTnet)

Table S4. ^{177}Lu-PSMA-617 quality control specifications

Test	Specification
Validation testing QC*	
Visual inspection	Clear, colourless to slightly yellow, without visible particles
Identity (HPLC)	confirms to retention time of reference standard ± 0.5 min
Half-life Lu-177	6.65 days ± 5 % (Dose calibrator)
Radiochemical purity (TLC)*	≥95.0 (at t = 0) >95.0 (at t = 4 hrs or nominated shelf life post EOS)
Radiochemical purity (HPLC)	>90.0 % (at t = 0) >90.0 % (at t = 4 hrs or nominated shelf life post EOS)
Lu-PSMA-617 + PSMA-617 with regard to overall impurities (100 %) within a range of retention time 8.00 to 15.00 min (HPLC (UV)) (optional)	Σ ≥ 90.0 % (HPLC, UV)
PSMA-617 (optional)	≤ 15.0 µg/MBq (HPLC, UV)
Lu–PSMA-617 (optional)	≤ 15.0 µg/MBq (HPLC, UV)
[^{177}Lu]Lu –PSMA-617	≥ 90.0 % (calculated as difference from 100 % and the sum of the test results (in %) of "radiochemical impurities" (HPLC (radio))
pH	4.5 – 7.5 (pH indicator stripes)
filter integrity	passed test (Bubble point test)
Bacterial endotoxins	≤ 175 IU/V (Ph.Eur. 2.6.14, method D)

Sterility (post release test) (if performed)	Sterile (Ph.Eur. 2.6 1)
Batch release QC (for each synthesis)	
Visual inspection	Clear, colourless to slightly yellow, without visible particles
Label inspection	Product, volume, activity, date, time
pH	4.5 – 7.5
Radiochemical purity (TLC)	≥95.0%
Radiochemical purity (HPLC) (optional)	≥90.0 %
Identity	confirms to retention time of reference standard ± 0.5 min

*three consecutive successful validations required. PET camera site validation also undertaken by the Australasian Radiopharmaceutical Trials network (ARTnet)

Table S5. Baseline characteristics

Characteristic		Cabazitaxel	Lu-PSMA
Age (Years):	Missing	0	0
[N] Mean (SD)		[101] 71.5 (7)	[99] 71.7 (7.9)
Median (IQR)		71.8 (66.7 to 77.3)	72.1 (66.9 to 76.7)
Min to Max		51.2 to 84.3	49.5 to 85.6
Weight (kg):	Missing	1	0
[N] Mean (SD)		[100] 87 (15.5)	[99] 92 (17.8)
Median (IQR)		84 (78 to 95)	88 (77 to 108)
Min to Max		55 to 140	57 to 158
BSA m^2:	Missing	3	0
[N] Mean (SD)		[98] 2 (0.2)	[99] 2.1 (0.2)
Median (IQR)		2 (1.9 to 2.1)	2.1 (1.9 to 2.2)
Min to Max		1.6 to 2.5	1.6 to 2.7
Gleason Score Category:	Missing	16 (15.8%)	21 (21.2%)

Characteristic		Cabazitaxel	Lu-PSMA
	8-10	50 (49.5%)	53 (53.5%)
	<=7	35 (34.7%)	25 (25.3%)
Prior curative intent radiotherapy?	No	55 (54.5%)	59 (59.6%)
	Yes	46 (45.5%)	40 (40.4%)
Prior radical prostatectomy?	No	57 (56.4%)	56 (56.6%)
	Yes	44 (43.6%)	43 (43.4%)
Visceral Mets (lung, liver, other)	No	88 (87.1%)	92 (92.9%)
	Yes	13 (12.9%)	7 (7.1%)
Lymph node metastases	No	54 (53.5%)	47 (47.5%)
	Yes	47 (46.5%)	52 (52.5%)
Bone metastases	No	11 (10.9%)	9 (9.1%)
	Yes	90 (89.1%)	90 (90.9%)
Sites of disease based on central review	20 or less	22 (21.8%)	22 (22.2%)
	Greater than 20	79 (78.2%)	77 (77.8%)
Prior treatment with abiraterone or enzalutamide	Abiraterone only	24 (23.8%)	21 (21.2%)
	Both Enzalutamide and Abiraterone	9 (8.9%)	21 (21.2%)
	Enzalutamide only	58 (57.4%)	49 (49.5%)
	Neither Enzalutamide nor Abiraterone	10 (9.9%)	8 (8.1%)
ECOG performance status	0	44 (43.6%)	42 (42.4%)
	1	52 (51.5%)	53 (53.5%)

Characteristic		Cabazitaxel	Lu-PSMA
	2	4 (4.0%)	4 (4.0%)
	Missing	1 (1.0%)	
Baseline PSA: Missing		0	0
[N] Mean (SD)		[101] 266.4 (450.8)	[99] 268.9 (681.2)
Median (IQR)		110 (64.2 to 245)	93.5 (44 to 219)
Min to Max		15.1 to 2730	8 to 6230
Platelets (10^9/L): Missing		0	0
[N] Mean (SD)		[101] 262.4 (85.4)	[99] 238.7 (75.6)
Median (IQR)		241 (209 to 308)	229 (180 to 275)
Min to Max		103 to 550	100 to 509
White cell count (10^9/L): Missing		0	0
[N] Mean (SD)		[101] 7.1 (2.4)	[99] 6.7 (2.1)
Median (IQR)		6.7 (5.3 to 8.2)	6.4 (5.4 to 7.5)
Min to Max		3.8 to 18.4	3.1 to 17.3
Haemoglobin (g/L): Missing		0	0
[N] Mean (SD)		[101] 128.3 (14.3)	[99] 125 (12.3)
Median (IQR)		128 (122 to 137)	124 (117 to 133)
Min to Max		88 to 163	99 to 155
ALP (U/L): Missing		0	0
[N] Mean (SD)		[101] 206.6 (267.9)	[99] 175.7 (172.2)
Median (IQR)		130 (79 to 187)	111 (83 to 199)
Min to Max		31 to 1947	35 to 1070
LDH (U/L): Missing		44	49

Characteristic	Cabazitaxel	Lu-PSMA
[N] Mean (SD)	[57] 271.5 (125.2)	[50] 275.4 (120.7)
Median (IQR)	237 (198 to 278)	239.5 (202 to 302)
Min to Max	135 to 800	156 to 772

Table S6. Outcomes of six exceptional responders who received further on-study ^{177}Lu-PSMA-617 at time of progression

Pt No	Time to retreatment from last cycle (days)	Time to retreatment from rando (days)	PSA response >50% from retreatment	Any PSA response from retreatment	Time from rando to 1st PSA progression (days)	Time from rando to 2nd PSA progression (days)	Time from retreatment to 2nd PSA progression (days)
1	309	497	Yes	Yes	421	596	99
2	399	581	No	Yes	485	722	141
3	553	616	Yes	Yes	260	Ongoing PSA response	Ongoing PSA response
4	168	232	No	No	79	292	60
5	211	265	No	No	210	307	42
6	105	202	Yes	Yes	155	348	146

Table S7. Progression free survival per protocol analysis

	PFS (95%CI)	rPFS (95%CI)	PSA-PFS (95%CI)
Rate at 12 months			
^{177}Lu-PSMA-617	19% (12-27%)	30% (21-39%)	21% (13-29%)
cabazitaxel	3% (1-9%)	14% (8-23%)	4% (1-10%)
HR	0.60 (0.44 - 0.82)	0.63 (0.46 - 0.87)	0.58 (0.42 - 0.80)
P value	0.001	0.005	<0.0001

Table S8. All adverse events (any cause)

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Blood and lymphatic system disorders: total	9	6	17	1	_	33	13	14	12	1	_	40	73
Anemia	8	3	7	_	_	18	8	11	8	_	_	27	45
Neutropenia (+/- fever)	1	3	10	1	_	15	4	3	3	1	_	11	26
Thrombotic thrombocytopenic purpura	_	_	_	_	_	_	1	_	1	_	_	2	2
Cardiac disorders: total	1	2	3	1	_	7	4	5	3	_	_	12	19
Acute coronary syndrome	_	_	_	_	_	_	_	1	1	_	_	2	2
Atrial fibrillation	_	1	2	_	_	3	2	2	_	_	_	4	7
Heart failure	_	_	_	_	_	_	_	_	1	_	_	1	1
Myocardial infarction	_	_	1	1	_	2	_	_	1	_	_	1	3
Palpitations	_	1	_	_	_	1	_	_	_	_	_	_	1
Sinus tachycardia	1	_	_	_	_	1	2	1	_	_	_	3	4
Supraventricular tachycardia	_	_	_	_	_	_	_	1	_	_	_	1	1
Ear and labyrinth disorders: total	2	_	1	_	_	3	4	_	1	_	_	5	8
Hearing impaired	1	_	_	_	_	1	1	_	_	_	_	1	2
Tinnitus	1	_	_	_	_	1	2	_	_	_	_	2	3
Vertigo	_	_	1	_	_	1	1	_	1	_	_	2	3
Endocrine disorders: total	_	_	_	_	_	_	_	2	_	_	_	2	2
Adrenal insufficiency	_	_	_	_	_	_	_	2	_	_	_	2	2
Eye disorders: total	7	_	_	_	_	7	26	4	_	_	_	30	37
Conjunctivitis	1	_	_	_	_	1	_	1	_	_	_	1	2
Dry eye	3	_	_	_	_	3	26	3	_	_	_	29	32
Eye disorders - Other, specify	1	_	_	_	_	1	_	_	_	_	_	_	1
Eye pain	1	_	_	_	_	1	_	_	_	_	_	_	1
Watering eyes	1	_	_	_	_	1	_	_	_	_	_	_	1
Gastrointestinal disorders: total	130	31	6	_	_	167	150	45	5	_	_	200	367
Abdominal distension	1	_	_	_	_	1	_	_	_	_	_	_	1
Abdominal pain	3	1	_	_	_	4	3	1	_	_	_	4	8
Anal hemorrhage	1	_	_	_	_	1	_	1	_	_	_	1	2
Bloating	1	_	_	_	_	1	1	1	_	_	_	2	3
Colonic hemorrhage	_	1	_	_	_	1	_	_	_	_	_	_	1
Constipation	17	8	_	_	_	25	28	10	_	_	_	38	63

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Dental caries	_	_	_	_	_	_	_	1	_	_	_	1	1
Diarrhea	35	9	4	_	_	48	15	3	1	_	_	19	67
Dry mouth	15	3	_	_	_	18	45	14	_	_	_	59	77
Dyspepsia	_	_	_	_	_	_	2	_	_	_	_	2	2
Dysphagia	1	1	_	_	_	2	1	1	_	_	_	2	4
Enterocolitis	_	_	_	_	_	_	_	_	1	_	_	1	1
Fecal incontinence	2	_	_	_	_	2	1	_	_	_	_	1	3
Gastritis	1	_	_	_	_	1	_	_	_	_	_	_	1
Gastroesophageal reflux disease	7	_	_	_	_	7	4	_	_	_	_	4	11
Gastrointestinal disorders - Other, specify	2	_	_	_	_	2	3	1	_	_	_	4	6
Gastrointestinal pain	1	_	_	_	_	1	_	_	_	_	_	_	1
Ileal obstruction	_	_	_	_	_	_	_	_	1	_	_	1	1
Mucositis oral	7	_	_	_	_	7	3	_	_	_	_	3	10
Nausea	24	5	_	_	_	29	31	8	1	_	_	40	69
Oral pain	_	_	_	_	_	_	1	1	_	_	_	2	2
Periodontal disease	1	_	_	_	_	1	1	_	_	_	_	1	2
Rectal hemorrhage	2	_	_	_	_	2	_	_	_	_	_	_	2
Rectal pain	1	_	_	_	_	1	_	_	_	_	_	_	1
Toothache	1	_	_	_	_	1	2	_	_	_	_	2	3
Vomiting	7	3	2	_	_	12	9	3	1	_	_	13	25
General disorders and administration site conditions: total	78	50	7	_	_	135	85	50	13	_	2	150	285
Edema face	_	_	_	_	_	_	_	1	_	_	_	1	1
Edema limbs	6	5	_	_	_	11	8	2	1	_	_	11	22
Fatigue	41	20	3	_	_	64	46	23	5	_	_	74	138
Fever	6	2	1	_	_	9	3	3	_	_	1	7	16
Flu like symptoms	1	_	_	_	_	1	6	_	_	_	_	6	7
Gait disturbance	_	2	_	_	_	2	_	_	_	_	_	_	2
General disorders and administration site conditions - Other, specify	_	1	_	_	_	1	_	_	1	_	1	2	3
Infusion related reaction	_	7	_	_	_	7	_	_	_	_	_	_	7
Irritability	1	_	_	_	_	1	_	_	_	_	_	_	1
Localized edema	_	_	_	_	_	_	2	_	_	_	_	2	2
Malaise	1	_	_	_	_	1	_	_	_	_	_	_	1

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Non-cardiac chest pain	1	1	1	_	_	3	_	_	_	_	_	_	3
Pain	21	12	2	_	_	35	20	21	6	_	_	47	82
Hepatobiliary disorders: total	_	_	_	_	_	_	_	_	2	_	1	3	3
Bile duct stenosis	_	_	_	_	_	_	_	_	1	_	_	1	1
Gallbladder obstruction	_	_	_	_	_	_	_	_	1	_	_	1	1
Hepatic failure	_	_	_	_	_	_	_	_	_	_	1	1	1
Immune system disorders: total	_	3	_	_	_	3	_	1	_	_	_	1	4
Allergic reaction	_	3	_	_	_	3	_	_	_	_	_	_	3
Immune system disorders - Other, specify	_	_	_	_	_	_	_	1	_	_	_	1	1
Infections and infestations: total	4	19	8	1	2	34	5	14	12	3	2	36	70
Bladder infection	_	1	_	_	_	1	_	_	_	_	_	_	1
Bone infection	_	_	_	_	_	_	_	_	1	_	_	1	1
Bronchial infection	_	1	1	_	_	2	_	1	2	_	_	3	5
Infections and infestations - Other, specify	2	1	_	1	_	4	3	1	_	_	_	4	8
Joint infection	_	_	_	_	_	_	_	1	_	_	_	1	1
Lung infection	_	2	2	_	1	5	_	2	3	1	1	7	12
Mucosal infection	1	_	_	_	_	1	_	_	_	_	_	_	1
Penile infection	_	_	_	_	_	_	_	1	_	_	_	1	1
Scrotal infection	_	_	_	_	_	_	_	_	1	_	_	1	1
Sepsis	_	_	_	_	1	1	_	_	_	2	1	3	4
Skin infection	1	1	1	_	_	3	1	_	_	_	_	1	4
Upper respiratory infection	_	9	_	_	_	9	1	4	2	_	_	7	16
Urinary tract infection	_	4	4	_	_	8	_	4	3	_	_	7	15
Injury, poisoning and procedural complications: total	5	4	1	_	_	10	3	8	4	_	1	16	26
Aortic injury	_	_	_	_	_	_	_	_	_	_	1	1	1
Bruising	_	_	_	_	_	_	2	_	_	_	_	2	2
Dermatitis radiation	1	_	_	_	_	1	_	_	_	_	_	_	1
Fall	3	1	1	_	_	5	_	5	2	_	_	7	12
Fracture	_	_	_	_	_	_	_	1	1	_	_	2	2
Injury, poisoning and procedural complications - Other, specify	_	1	_	_	_	1	_	1	_	_	_	1	2

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Rectal anastomotic leak	_	1	_	_	_	1	_	_	_	_	_	_	1
Spinal fracture	_	1	_	_	_	1	1	1	1	_	_	3	4
Wrist fracture	1	_	_	_	_	1	_	_	_	_	_	_	1
Investigations: total	14	11	2	1	_	28	30	18	10	4	_	62	90
Alanine aminotransferase increased	_	2	_	_	_	2	_	1	1	_	_	2	4
Alkaline phosphatase increased	1	1	1	_	_	3	3	_	_	_	_	3	6
Aspartate aminotransferase increased	_	1	_	_	_	1	1	_	1	_	_	2	3
Blood bilirubin increased	_	_	_	_	_	_	1	1	_	_	_	2	2
Creatinine increased	1	1	_	_	_	2	3	1	_	_	_	4	6
GGT increased	3	_	1	_	_	4	_	_	_	_	_	_	4
Investigations - Other, specify	_	_	_	_	_	_	1	1	_	_	_	2	2
Lymphocyte count decreased	1	1	_	_	_	2	2	1	_	_	_	3	5
Platelet count decreased	2	2	_	_	_	4	8	10	7	4	_	29	33
Weight gain	3	_	_	_	_	3	_	_	_	_	_	_	3
Weight loss	1	2	_	_	_	3	6	1	_	_	_	7	10
White blood cell decreased	2	1	_	1	_	4	5	2	1	_	_	8	12
Metabolism and nutrition disorders: total	24	5	3	1	_	33	29	11	1	_	_	41	74
Anorexia	17	5	1	_	_	23	22	4	_	_	_	26	49
Dehydration	_	_	_	_	_	_	1	1	1	_	_	3	3
Hypercalcemia	_	_	_	1	_	1	1	1	_	_	_	2	3
Hyperglycemia	1	_	_	_	_	1	1	_	_	_	_	1	2
Hyperkalemia	_	_	_	_	_	_	1	_	_	_	_	1	1
Hypermagnesemia	_	_	_	_	_	_	1	_	_	_	_	1	1
Hypoalbuminemia	_	_	_	_	_	_	_	1	_	_	_	1	1
Hypocalcemia	2	_	_	_	_	2	1	_	_	_	_	1	3
Hypokalemia	1	_	1	_	_	2	_	1	_	_	_	1	3
Hypomagnesemia	2	_	_	_	_	2	_	1	_	_	_	1	3
Hyponatremia	_	_	1	_	_	1	_	_	_	_	_	_	1
Hypophosphatemia	_	_	_	_	_	_	1	2	_	_	_	3	3
Metabolism and nutrition disorders - Other, specify	1	_	_	_	_	1	_	_	_	_	_	_	1
Musculoskeletal and connective tissue disorders: total	47	20	5	_	_	72	42	33	6	_	_	81	153

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Arthralgia	3	_	_	_	_	3	1	1	_	_	_	2	5
Arthritis	1	_	_	_	_	1	2	3	1	_	_	6	7
Back pain	12	11	_	_	_	23	16	10	3	_	_	29	52
Bone pain	11	2	_	_	_	13	7	8	1	_	_	16	29
Buttock pain	1	_	_	_	_	1	_	_	_	_	_	_	1
Chest wall pain	2	1	_	_	_	3	1	1	_	_	_	2	5
Flank pain	1	2	_	_	_	3	_	1	_	_	_	1	4
Generalized muscle weakness	1	_	_	_	_	1	_	_	_	_	_	_	1
Muscle weakness lower limb	2	2	3	_	_	7	_	_	_	_	_	_	7
Muscle weakness upper limb	2	_	_	_	_	2	_	_	_	_	_	_	2
Musculoskeletal and connective tissue disorder - Other, specify	2	_	_	_	_	2	2	2	_	_	_	4	6
Myalgia	4	1	_	_	_	5	1	_	_	_	_	1	6
Neck pain	1	_	_	_	_	1	2	2	_	_	_	4	5
Osteonecrosis of jaw	_	_	_	_	_	_	1	_	_	_	_	1	1
Osteoporosis	_	_	1	_	_	1	1	_	_	_	_	1	2
Pain in extremity	4	1	1	_	_	6	8	5	1	_	_	14	20
Neoplasms benign, malignant and unspecified (incl cysts and polyps): total	2	2	2	_	4	10	_	4	_	2	7	13	23
Neoplasms benign, malignant and unspecified (incl cysts and polyps) - Other, specify	2	1	2	_	4	9	_	2	_	2	7	11	20
Tumor pain	_	1	_	_	_	1	_	2	_	_	_	2	3
Nervous system disorders: total	54	23	6	_	_	83	44	10	3	_	_	57	140
Dizziness	8	3	_	_	_	11	4	_	_	_	_	4	15
Dysarthria	_	_	_	_	_	_	1	_	_	_	_	1	1
Dysgeusia	16	7	_	_	_	23	11	1	_	_	_	12	35
Facial nerve disorder	_	_	_	_	_	_	1	_	_	_	_	1	1
Headache	5	1	_	_	_	6	5	1	_	_	_	6	12
Hypersomnia	_	_	_	_	_	_	1	_	_	_	_	1	1
Intracranial hemorrhage	_	_	1	_	_	1	_	_	_	_	_	_	1
Lethargy	2	_	_	_	_	2	5	1	_	_	_	6	8
Memory impairment	2	_	_	_	_	2	1	_	_	_	_	1	3
Nervous system disorders - Other, specify	1	1	_	_	_	2	_	1	1	_	_	2	4

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Neuralgia	3	_	_	_	_	3	1	_	_	_	_	1	4
Paresthesia	3	3	_	_	_	6	5	1	_	_	_	6	12
Peripheral neuropathy	14	8	1	_	_	23	8	2	_	_	_	10	33
Presyncope	_	_	_	_	_	_	_	3	_	_	_	3	3
Pyramidal tract syndrome	_	_	_	_	_	_	_	_	1	_	_	1	1
Seizure	_	_	_	_	_	_	_	_	1	_	_	1	1
Somnolence	_	_	_	_	_	_	1	_	_	_	_	1	1
Syncope	_	_	3	_	_	3	_	_	_	_	_	_	3
Vasovagal reaction	_	_	1	_	_	1	_	_	_	_	_	_	1
Psychiatric disorders: total	15	2	3	_	_	20	14	5	2	_	_	21	41
Agitation	_	_	_	_	_	_	1	_	_	_	_	1	1
Anxiety	_	1	_	_	_	1	_	2	_	_	_	2	3
Confusion	1	_	1	_	_	2	1	1	1	_	_	3	5
Delirium	_	_	_	_	_	_	_	_	1	_	_	1	1
Depression	3	_	1	_	_	4	5	_	_	_	_	5	9
Insomnia	11	1	1	_	_	13	7	2	_	_	_	9	22
Renal and urinary disorders: total	17	12	10	_	_	39	13	5	5	_	_	23	62
Acute kidney injury	_	2	1	_	_	3	_	_	1	_	_	1	4
Bladder spasm	_	_	_	_	_	_	_	1	_	_	_	1	1
Cystitis noninfective	_	_	1	_	_	1	_	1	_	_	_	1	2
Hematuria	10	2	5	_	_	17	3	_	1	_	_	4	21
Renal and urinary disorders - Other, specify	_	_	_	_	_	_	_	_	1	_	_	1	1
Renal calculi	_	_	_	_	_	_	_	_	1	_	_	1	1
Urinary frequency	4	2	_	_	_	6	3	_	_	_	_	3	9
Urinary incontinence	2	3	1	_	_	6	3	1	_	_	_	4	10
Urinary retention	_	3	1	_	_	4	_	2	_	_	_	2	6
Urinary tract obstruction	_	_	1	_	_	1	1	_	1	_	_	2	3
Urinary tract pain	1	_	_	_	_	1	3	_	_	_	_	3	4
Reproductive system and breast disorders: total	2	3	2	_	_	7	1	_	1	_	_	2	9
Erectile dysfunction	_	_	1	_	_	1	_	_	_	_	_	_	1
Genital edema	_	1	1	_	_	2	_	_	1	_	_	1	3
Gynecomastia	1	_	_	_	_	1	_	_	_	_	_	_	1
Pelvic pain	1	1	_	_	_	2	1	_	_	_	_	1	3

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Testicular pain	_	1	_	_	_	1	_	_	_	_	_	_	1
Respiratory, thoracic and mediastinal disorders: total	24	13	1	2	_	40	25	4	3	_	_	32	72
Allergic rhinitis	1	_	_	_	_	1	_	_	_	_	_	_	1
Cough	9	3	_	_	_	12	11	1	_	_	_	12	24
Dyspnea	8	8	1	_	_	17	9	2	3	_	_	14	31
Epistaxis	1	_	_	_	_	1	_	1	_	_	_	1	2
Hiccups	1	_	_	_	_	1	1	_	_	_	_	1	2
Laryngeal inflammation	1	_	_	_	_	1	_	_	_	_	_	_	1
Nasal congestion	2	_	_	_	_	2	_	_	_	_	_	_	2
Pleuritic pain	_	_	_	_	_	_	1	_	_	_	_	1	1
Pneumonitis	_	1	_	1	_	2	_	_	_	_	_	_	2
Pneumothorax	_	_	_	1	_	1	_	_	_	_	_	_	1
Productive cough	_	1	_	_	_	1	_	_	_	_	_	_	1
Pulmonary hypertension	_	_	_	_	_	_	1	_	_	_	_	1	1
Sore throat	_	_	_	_	_	_	2	_	_	_	_	2	2
Wheezing	1	_	_	_	_	1	_	_	_	_	_	_	1
Skin and subcutaneous tissue disorders: total	15	3	_	_	_	18	9	1	_	_	_	10	28
Alopecia	5	1	_	_	_	6	2	_	_	_	_	2	8
Body odor	1	_	_	_	_	1	_	_	_	_	_	_	1
Dry skin	1	_	_	_	_	1	1	_	_	_	_	1	2
Erythema multiforme	_	_	_	_	_	_	1	_	_	_	_	1	1
Hyperhidrosis	_	_	_	_	_	_	1	_	_	_	_	1	1
Palmar-plantar erythrodysesthesia syndrome	_	1	_	_	_	1	_	_	_	_	_	_	1
Pruritus	1	_	_	_	_	1	_	1	_	_	_	1	2
Rash acneiform	1	_	_	_	_	1	_	_	_	_	_	_	1
Rash maculo-papular	2	_	_	_	_	2	1	_	_	_	_	1	3
Skin and subcutaneous tissue disorders - Other, specify	3	_	_	_	_	3	2	_	_	_	_	2	5
Skin ulceration	1	1	_	_	_	2	1	_	_	_	_	1	3
Social circumstances: total	_	_	1	_	_	1	_	1	_	_	_	1	2
Social circumstances - Other, specify	_	_	1	_	_	1	_	1	_	_	_	1	2
Surgical and medical procedures: total	_	_	1	_	_	1	_	_	_	_	_	_	1

Class / Term	Cabazitaxel						Lu-PSMA						Total
	G1	G2	G3	G4	G5	N	G1	G2	G3	G4	G5	N	
Surgical and medical procedures - Other, specify	_	_	1	_	_	1	_	_	_	_	_	_	1
Vascular disorders: total	14	6	3	_	_	23	8	4	3	1	_	16	39
Flushing	2	1	_	_	_	3	_	_	_	_	_	_	3
Hot flashes	8	_	_	_	_	8	3	1	_	_	_	4	12
Hypertension	3	1	1	_	_	5	1	_	1	_	_	2	7
Hypotension	1	2	_	_	_	3	1	_	_	_	_	1	4
Lymphedema	_	_	_	_	_	_	1	_	_	_	_	1	1
Thromboembolic event	_	2	2	_	_	4	_	2	2	1	_	5	9
Vascular disorders - Other, specify	_	_	_	_	_	_	2	_	_	_	_	2	2
Vasculitis	_	_	_	_	_	_	_	1	_	_	_	1	1

Table S9. Reasons for treatment discontinuation

	Lu-PSMA	Cabazitaxel
Completion of all planned treatment	38	29
Disease progression	31	25
Death	9	2
Adverse event	1	3
Other	19	26
Total	**98**	**85**

Table S10. Reasons for dose reduction

Treatment	Event No	Reason for dose reduction
LuPSMA	1	Creatinine increased
	2	Diarrhoea, vomiting, anemia
	3	Dry mouth
	4	Fatigue
	5	Lutetium availability
	6	Nausea
	7	Neutropenia
	8	Preparation dose reduction
	9	Shortness of breath
	10	Thrombocytopenia
	11	Thrombocytopenia
	12	Thrombocytopenia, neutropenia

Cabazitaxel	1	Anaemia, diarrhoea, hypercalcaemia
	2	Change in patient weight
	3	Diarrhoea
	4	Diarrhoea
	5	Fatigue
	6	Fatigue
	7	Fatigue
	8	Febrile neutropenia
	9	Febrile neutropenia
	10	Foot ulcer
	11	Hypotension
	12	Infusion reaction, nausea, diarrhoea
	13	Nausea
	14	Neutropenia
	15	Peripheral neuropathy
	16	Peripheral neuropathy
	17	Peripheral neuropathy
	18	Pneumonitis
	19	Thrombocytopenia
	20	Urinary tract infection
	21	Vomiting

Table S11. Summary of EORTC QLQ-C30 repeated measures analysis

Scale	Cabazitaxel	Lu-PSMA	Diff.
	Pred. Mean (SE) {95% CI}	Pred. Mean (SE) {95% CI}	Pred. Mean (SE) {95% CI} [p-value]
Physical Functioning	73.0 (1.5) {70.0 to 75.9}	74.2 (1.3) {71.6 to 76.9}	1.3 (2.0) {-2.6 to 5.1} [0.526]
Role Functioning	67.0 (2.0) {63.0 to 71.0}	69.9 (1.8) {66.4 to 73.5}	2.9 (2.6) {-2.3 to 8.1} [0.275]
Emotional Functioning	79.7 (1.4) {76.8 to 82.5}	82.1 (1.3) {79.6 to 84.6}	2.4 (1.9) {-1.3 to 6.1} [0.206]
Cognitive Functioning	82.4 (1.4) {79.6 to 85.2}	84.2 (1.3) {81.7 to 86.7}	1.8 (1.9) {-1.9 to 5.5} [0.332]
Social Functioning	73.0 (1.9) {69.2 to 76.9}	78.6 (1.7) {75.2 to 81.9}	5.5 (2.5) {0.5 to 10.5} [0.030]
Global health status / QoL	60.4 (1.8) {56.9 to 63.9}	63.4 (1.6) {60.3 to 66.5}	3.0 (2.3) {-1.6 to 7.5} [0.202]
Fatigue	39.6 (1.8) {36.0 to 43.1}	34.3 (1.6) {31.2 to 37.5}	-5.3 (2.4) {-9.9 to -0.6} [0.027]

Scale	Cabazitaxel	Lu-PSMA	Diff.
	Pred. Mean (SE) {95% CI}	Pred. Mean (SE) {95% CI}	Pred. Mean (SE) {95% CI} [p-value]
Nausea / Vomiting	9.3 (1.4) {6.5 to 12.1}	8.1 (1.3) {5.6 to 10.6}	-1.2 (1.9) {-4.9 to 2.5} [0.526]
Pain	26.9 (1.9) {23.0 to 30.7}	25.4 (1.7) {21.9 to 28.8}	-1.5 (2.5) {-6.5 to 3.5} [0.556]
Dyspnoea	23.3 (1.9) {19.6 to 27.1}	23.9 (1.7) {20.6 to 27.3}	0.6 (2.5) {-4.3 to 5.5} [0.816]
Insomnia	29.0 (2.0) {25.1 to 32.9}	23.1 (1.8) {19.6 to 26.6}	-5.9 (2.6) {-11.0 to -0.8} [0.023]
Appetite loss	20.2 (2.1) {16.0 to 24.4}	20.4 (1.9) {16.7 to 24.2}	0.3 (2.8) {-5.2 to 5.8} [0.919]
Constipation	12.6 (1.6) {9.4 to 15.8}	13.2 (1.4) {10.3 to 16.0}	0.5 (2.1) {-3.6 to 4.6} [0.796]
Diarrhoea	15.5 (1.5) {12.6 to 18.5}	8.6 (1.3) {5.9 to 11.2}	-7.0 (1.9) {-10.8 to -3.1} [0.000]
Financial Problems	12.5 (1.7) {9.2 to 15.8}	11.8 (1.5) {8.9 to 14.7}	-0.7 (2.2) {-5.0 to 3.6} [0.754]

Table S12. Frequency of troublesome symptoms

Scale	Lu-PSMA	Cabazitaxel	OR (95% CI)*	P-value*
01. Pain (all and anywhere)	69 (73.4%)	58 (78.4%)	0.61 (0.28 to 1.34)	0.217
02. Fatigue (tiredness)	77 (81.9%)	64 (87.7%)	0.65 (0.25 to 1.69)	0.379
03. Poor appetite	52 (55.3%)	43 (58.9%)	0.73 (0.36 to 1.5)	0.396
04. Cough	37 (39.4%)	35 (47.3%)	0.83 (0.42 to 1.61)	0.573
05. Shortness of breath	47 (50.5%)	37 (50.0%)	0.94 (0.42 to 2.07)	0.872
06. Trouble sleeping	56 (60.2%)	49 (67.1%)	0.78 (0.38 to 1.61)	0.498
07. Nausea	30 (32.3%)	27 (36.5%)	0.81 (0.42 to 1.56)	0.533
08. Vomiting	13 (14.0%)	18 (24.3%)	0.46 (0.2 to 1.03)	0.060
09. Diarrhoea	29 (31.2%)	41 (55.4%)	0.34 (0.18 to 0.66)	0.001
10. Constipation	51 (54.8%)	40 (54.1%)	1.06 (0.55 to 2.04)	0.859
11. Urinary symptoms	34 (36.6%)	45 (61.6%)	0.32 (0.16 to 0.66)	0.002
12. Leg swelling	26 (28.0%)	22 (29.7%)	0.91 (0.43 to 1.92)	0.804
13. Difficulty walking	56 (60.2%)	43 (58.9%)	0.82 (0.39 to 1.75)	0.614
14. Anxiety (feeling worried)	41 (43.6%)	32 (43.2%)	1.01 (0.5 to 2.02)	0.979

Scale	Lu-PSMA	Cabazitaxel	OR (95% CI)*	P-value*
15. Depression (feeling sad)	36 (38.7%)	32 (43.2%)	0.72 (0.35 to 1.51)	0.389
16. Irritability (being cranky)	48 (51.1%)	34 (45.9%)	1.14 (0.55 to 2.34)	0.731
17. Trouble concentrating	37 (39.4%)	39 (52.7%)	0.52 (0.25 to 1.06)	0.073
18. Energy (reversed)	88 (93.6%)	70 (94.6%)	0.84 (0.18 to 3.92)	0.824
19. Appetite (reversed)	76 (80.9%)	62 (83.8%)	0.7 (0.29 to 1.67)	0.417
20. Mobility (reversed)	78 (83.0%)	63 (85.1%)	0.72 (0.28 to 1.83)	0.486
21. Mood (reversed)	72 (77.4%)	58 (78.4%)	0.7 (0.27 to 1.79)	0.453
22. Physical well-being (reversed)	79 (84.0%)	64 (86.5%)	0.74 (0.27 to 2.01)	0.557
23. Emotional well-being (reversed)	70 (74.5%)	58 (78.4%)	0.65 (0.28 to 1.54)	0.329
24. Overall well-being (reversed)	75 (79.8%)	62 (83.8%)	0.68 (0.27 to 1.68)	0.399
25. Drowsiness	58 (61.7%)	53 (71.6%)	0.49 (0.23 to 1.04)	0.063
26. Headaches	20 (21.3%)	23 (31.1%)	0.49 (0.23 to 1.05)	0.068
27. Feeling dizzy or lightheaded	35 (38.0%)	39 (52.7%)	0.39 (0.19 to 0.79)	0.009
28. Dry mouth	65 (69.1%)	47 (63.5%)	1.34 (0.67 to 2.68)	0.404
29. Altered sense of taste	51 (54.3%)	51 (68.9%)	0.49 (0.25 to 0.98)	0.042
30. Sore mouth or throat	28 (29.8%)	32 (43.2%)	0.51 (0.26 to 1.03)	0.059
31. Difficulty swallowing	22 (23.4%)	19 (25.7%)	0.87 (0.41 to 1.84)	0.711
32. Hair loss	9 (9.6%)	31 (41.9%)	0.1 (0.04 to 0.26)	<.001
33. Skin rash	13 (13.8%)	24 (32.4%)	0.29 (0.13 to 0.66)	0.003
34. Sore hands and/or feet	33 (35.1%)	32 (43.2%)	0.43 (0.21 to 0.9)	0.024
35. Numbness or pins and needles	50 (53.8%)	44 (59.5%)	0.81 (0.41 to 1.62)	0.555
36. Fevers	16 (17.0%)	19 (25.7%)	0.44 (0.19 to 1.01)	0.052
37. Hot flashes	45 (47.9%)	40 (54.1%)	0.52 (0.24 to 1.13)	0.099
38. Sweating or sweats	50 (53.2%)	44 (59.5%)	0.55 (0.27 to 1.13)	0.105
39. Problems with sex	53 (64.6%)	47 (69.1%)	0.9 (0.39 to 2.11)	0.816
40. Problems taking tablets	12 (12.8%)	10 (13.5%)	1.02 (0.4 to 2.59)	0.971
41. Problems with needles or injections	10 (10.6%)	10 (13.5%)	0.9 (0.32 to 2.48)	0.833
42. Inconvenience of treatment	33 (35.1%)	31 (41.9%)	0.79 (0.4 to 1.57)	0.507
43. Problems coping with treatment	30 (32.3%)	28 (38.4%)	0.67 (0.34 to 1.31)	0.241
44. Thought of actually having treatment	22 (23.9%)	23 (31.1%)	0.81 (0.39 to 1.69)	0.576
45. Problems looking after myself	30 (31.9%)	23 (31.1%)	0.73 (0.34 to 1.56)	0.420
46. Problems doing what I wanted	54 (58.1%)	45 (61.6%)	0.86 (0.43 to 1.75)	0.681
47. Problems for my friends or family	42 (44.7%)	35 (47.9%)	0.71 (0.35 to 1.44)	0.340

*logistic regression adjusting for baseline value

Section S2: Clinical Protocol and Statistical Analysis Plan

Analysis Plan Changes Post Statistical Analysis Plan (SAP) Finalization

The trial management committee (TMC) made a change to the definition of the secondary progression free survival (PFS) endpoint after SAP approval. The decision was documented in the TMC meeting minutes on 02SEP20 and involved removing pain progression from the PFS endpoint. The rationale for the change was driven by doubt about reliability of a 1-point increase on the Present Pain Intensity (PPI) being truly indicative of a clinically meaningful increase in pain. The decision was made before any results were revealed to the TMC.

In collaboration with

Study Title

TheraP: A randomised phase 2 trial of ^{177}Lu-PSMA617 theranostic versus cabazitaxel in progressive metastatic castration resistant prostate cancer (ANZUP protocol 1603)

ANZUP protocol number: ANZUP 1603

Protocol version number and date: Version 2.0 30 July 2018

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Abbreviations

AE	Adverse Event
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
CT	Computed tomography (scan)
CRF	Case report form
CTC	NHMRC Clinical Trials Centre, University of Sydney
CTCAE	Common Terminology Criteria for Adverse Events
CR	Complete Response
mCRPC	Metastatic castration resistant prostate cancer
ECOG	Eastern Cooperative Oncology Group
EORTC	European Organisation for Research and Treatment of Cancer
EQ-5D-5L	Euroqol 5 item preference-based measure of health (5L)
^{18}F	Fluorine-18
FDG	Fluorodeoxyglucose
HBED	Hydroxybenzyl ethylenediamine
HREC	Human Research Ethics Committee
HRQL	Health-Related Quality of Life
IDSMC	Independent Data, Safety Monitoring Committee
INR	International Normalised Ratio
LHRHA	Luteinizing Hormone-Releasing Hormone analog
Lu-PSMA	^{177}Lu-PSMA 617
OS	Overall survival
OTRR	Objective tumour response rate
PCWG3	Prostate Cancer Working Group 3
PET	Positron emission tomography
PFS	Progression-Free Survival
PD	Progressive Disease
PDF	Patient Disease and Treatment Assessment Form
PPI	Present pain intensity (McGill-Melzack)
PR	Partial Response
PSMA	Prostate-specific membrane antigen
QLQ-C30	EORTC Core Quality of Life Questionnaire (30 items)
QOL	Quality of life
RECIST	Response Evaluation Criteria in Solid Tumours
SAE	Serious adverse event
SD	Stable Disease
SPECT	Single photon emission computed tomography
SUSAR	Suspected Unexpected Serious Adverse Reaction
SUV	Standardised uptake values
TGA	Therapeutic Goods Administration
ULN	Upper limit of normal
WBBS	Whole Body Bone Scan

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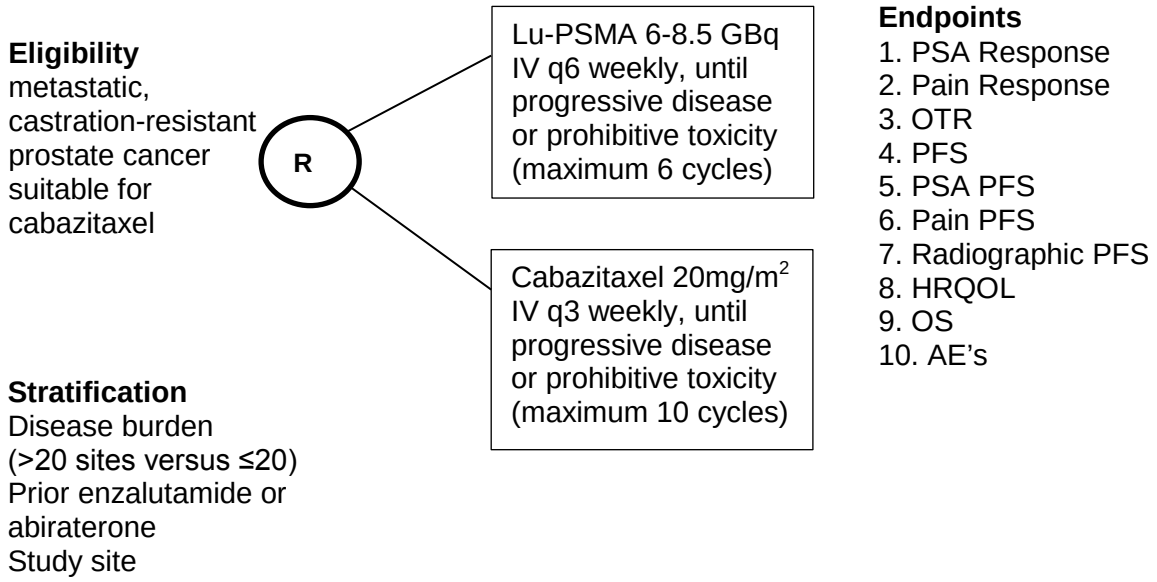
SYNOPSIS AND SCHEMA

Background	^{177}Lu-PSMA is a targeted radiopharmaceutical with promising activity and tolerability in single arm studies of men with metastatic prostate cancer progressing after multiple lines of chemotherapy and endocrine therapy. PSMA is a cell-surface protein overexpressed by the majority of prostate cancers. ^{177}Lutetium is a synthetic radioisotope with a half-life of ~7 days, and maximum tissue penetration of ~2 mm.
Aim	To determine the activity and safety of Lu-PSMA vs cabazitaxel
Objectives	To determine and compare the effects on
Primary	1. Prostate Specific Antigen (PSA) response rate (RR) (PSA reduction of ≥50% from baseline)
Secondary	2. Pain Response (PPI scale and analgesic score) 3. Objective tumour response (OTR = CR or PR, RECIST 1.1) rate 4. Progression-Free Survival (PFS = time to PSA progression, pain progression, radiographic progression, or death) 5. PSA progression-free survival (PCWG3) 6. Pain progression-free survival (PPI and analgesic score) 7. Radiographic progression-free survival (PCWG3 criteria for bone lesions and RECIST 1.1 for soft tissue lesions) 8. Aspects of Health related quality of life (HRQL, QLQ-C30, PDF) 9. Overall survival (death from any cause) 10. Frequency and severity of adverse events (CTCAE v 4.03)
	To determine
Tertiary correlative	11. Associations between ^{68}Ga-PSMA PET/CT, FDG-PET, baseline characteristics, and outcomes 12. Associations between clinical outcomes and possible prognostic and/or predictive biomarkers including ctDNA 13. Treatment and outcomes of participants excluded on basis of low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET
Design	Open label, randomised, stratified, 2-arm, multicentre, phase 2 trial
Population	Men with castration resistant prostate cancer suitable for chemotherapy with cabazitaxel (surgical or medical castration, and previous chemotherapy with docetaxel. Previous enzalutamide and/or abiraterone is permitted), ECOG performance status 0-2. ^{68}Ga-PSMA PET/CT must show high PSMA avidity without discordant disease on FDG PET/CT.
Study treatments	<u>Experimental</u>: Lu-PSMA 6-8.5 GBq IV every 6 weeks until prohibitive toxicity, the patient is no longer thought to be benefiting, or post-therapy imaging with SPECT/CT shows complete resolution of PSMA uptake, to a maximum of 6 cycles. Treatment can be interrupted after an exceptional response, re-started at subsequent progression, and then continued to a maximum of 6 cycles in total. <u>Standard</u>: Cabazitaxel 20mg/m^2 IV every 3 weeks with prednisolone 10mg daily orally, continued until progressive disease or prohibitive toxicity to a maximum of 10 cycles.
Assessments	Clinical assessments, HRQL, and PSA every 3 weeks during study treatment, then every 4 weeks until radiological progression.

	Imaging with CT and bone scan every 12 weeks. Imaging with ^{68}Ga-PSMA and FDG PET/CT at baseline, and PSMA SPECT 24 hours after each treatment with Lu-PSMA.
Statistical considerations	200 participants randomised in 1:1 ratio provide 80% power to detect a true absolute difference of 20% in the PSA RR from 40% in those allocated cabazitaxel, to 60% in those allocated lutetium PSMA, with a 2-sided type 1 error of 5% and an allowance of 3% for ineligible and/or inevaluable participants. This sample size also provides 80% power to detect a true hazard ratio for PFS of 0.65 assuming a median progression-free survival of 3 months in those allocated cabazitaxel, 24 months for accrual, an additional 6 months for follow-up, and a 2-sided type 1 error rate of 5%. The study has 80% power to detect a true hazard ratio for OS of 0.65 after 170 events.

Study Schema



1 BACKGROUND

1.1 Prostate cancer

Globally prostate cancer is the second most frequently diagnosed cancer in men and is a major cause of mortality leading to 258,000 deaths worldwide in 2008 (1). Although localised prostate cancer may be cured with radiotherapy or surgery, many patients will go on to develop metastatic disease (2). Standard initial treatment for patients with metastatic disease is androgen deprivation therapy (ADT) using either LHRH agonists / antagonists or anti-androgens such as bicalutamide that competitively block the androgen receptor. Although highly effective therapy, the median duration of response to medical castration is only 2 years and patients eventually develop resistance leading to disease progression and metastatic castration resistant prostate cancer (mCRPC).

1.2 Current treatment approaches for mCRPC

New therapies targeting the androgen receptor signalling axis, such as abiraterone and enzalutamide (3-6), and the taxanes docetaxel and cabazitaxel (7-9), all improve survival in men with mCRPC. Other agents with evidence for improved survival include Sipuleucel-T (10) and in bone only disease the α emitting radionuclide ^{223}Radium dichloride (11). Sipuleucel-T is not available in Australia and ^{223}Radium dichloride is currently not reimbursed in Australia.

1.3 Radionuclide therapy in prostate cancer

Prostate cancer is a radioresponsive disease and external beam radiotherapy is a proven curative modality for the treatment of localised prostate cancer. It is also a highly effective palliative therapy for localised metastatic disease. Radionuclide therapy uses systemically administered tumour-targeting agents to deliver a therapeutic radionuclide and provides another avenue for the application of radiation in treatment of metastatic prostate cancer. Radionuclide therapies that target the matrix surrounding bone metastases using both β and α emitting radionuclides have been shown to be effective therapies in metastatic prostate cancer (11, 12). However around a third of men with mCRPC present with either visceral or nodal metastases that cannot respond to therapies targeting bone metastases, limiting their more general applicability (13).

1.4 Prostate Specific Membrane Antigen (PSMA) as a target for radionuclide therapy

The ideal target for radionuclide therapy of prostate cancer would target tumour cells directly, and also target all sites of disease, not just bone. A number of cell surface markers have been identified (14, 15) and of these the most promising is PSMA.

PSMA, also known as folate hydrolase I, or glutamate carboxypeptidase II, is a 750-amino acid type II transmembrane glycoprotein expressed in normal human prostate epithelium that functions as a membrane metalloenzyme. In the central nervous system, PSMA catabolises N-acetylaspartyl glutamate (NAAG) to form N-acetyl aspartate and glutamate, which are selectively transported into glial cells where they are metabolised and re-used for NAAG synthesis in neurons. In the proximal jejunum PSMA is expressed on the plasma membrane of enterocytes where it sequentially hydrolyses the C-terminal glutamate of dietary folates leading to the release of folate-monoglutamate that can be absorbed into the bloodstream (16).

PSMA is over-expressed in virtually all prostate cancers and its expression is increased in poorly differentiated, metastatic and castration-resistant carcinomas (17, 18). The expression of PSMA in non-prostate tissues is limited to the small intestine, proximal renal tubules and lacrimal and salivary glands. In these tissues PSMA is expressed at levels 100-1000 times less than in the prostate (19, 20).

Following ligand binding to PSMA hetero-dimerisation occurs and is followed by rapid internalisation of the ligand receptor complex (21) . Following internalisation there is subsequent

transport to a peri-nuclear location (22, 23). This provides a mechanism for enhanced tumour uptake and retention of therapeutic radioisotope.

1.5 PSMA targeting strategies

A variety of different vectors are available to target PSMA with the two most developed approaches being monoclonal antibodies and small molecule inhibitors that target the macromolecular structure of PSMA and its enzymatic site respectively. Both approaches allow excellent tumour targeting and are rapidly internalised following ligand binding (24, 25). Antibody directed therapy of PSMA initially used the intact monoclonal antibody 7E11. This antibody binds to an intracellular portion of PSMA (26) and can only bind to necrotic cells. It demonstrated limited clinical efficacy for therapy and led to dose limiting haematological toxicity (27).

More recent work has focused on J591, a fully humanised monoclonal antibody that binds with high specificity to the extracellular domain of PSMA (17, 28). Radiolabelled J591 has been investigated in four early phase studies in men with mCRPC labelled with either ^{90}Y or ^{177}Lu (29-32). Modest anti-tumour effects and dose-limiting haematological toxicity have been observed. Efficacy in radio-immunotherapy using intact monoclonal antibodies is limited by their slow accumulation into solid tumour masses, with maximum uptake not seen for many days following therapy. Haematological toxicity results also relates to their long circulating half-life (33).

The active enzymatic site of PSMA is highly conserved and provides a robust target for the development of targeting agents (34). Recognising the biological function of PSMA there has been extensive research efforts developing PSMA inhibitors for a variety of therapeutic applications (35). The enzymatic site of PSMA can bind to other molecules that possess a similar structure to native substrate and a number of PSMA inhibitors have been developed. Small peptide inhibitors of PSMA, due to their small size, have both high binding affinity for PSMA resulting in rapid tumour uptake and high tumour absorbed dose, and also clear the circulation rapidly via the kidneys thereby reducing the risk of haematological toxicity.

To date the reported clinical experience using therapeutic radiolabelled small molecules targeting PSMA has been limited to small, retrospective institutional series (36-43). A number of urea based small molecules binding PSMA have been developed (44-49). ^{123}I labelled MIP-1072 and 1095 demonstrated excellent tumour localisation and clearance in first in man studies (48). ^{124}I and ^{131}I labelled MIP-1095 was utilised by Zechmann et al and heralded their potential as therapeutic avenue in prostate cancer (36). However the most developed agents are PSMA I&T (49) and PSMA-617 (45).

Experience in treating neuroendocrine tumours with β radiation-labelled somatostatin analogues, also small peptides, has shown that renal toxicity can largely be avoided when using shorter-range isotopes. Radiation dosimetry studies indicate that the short-range of radiation emitted from isotopes such as ^{177}Lu limits the absorbed dose received by the radiosensitive glomeruli. With use of ^{177}Lu-labelled somatostatin analogues reabsorption of the peptide necessitates co-administration of radioprotective amino acid infusion which promotes radiotracer excretion. Repeated administration of ^{177} Lu labelled somatostatin analogues results in minimal loss of renal function in patients treated for neuroendocrine tumours (50, 51). Radiolabelled small molecule inhibitors targeting PSMA, however, are not reabsorbed and therefore an amino acid infusion is not required. Renal dose with ^{177}Lu-PSMA617 are comparatively lower than ^{177}Lu-DOTATATE.

As a specific short-range therapeutic radionuclide, ^{177}Lu also has other favourable physical characteristics including a small γ radiation emission that allows for activity measurement ex vivo using SPECT-CT. It is also a radiometal and therefore is highly retained within cells following internalisation, a process referred to as residualisation.

1.6 Gallium-68 (^{68}Ga) PSMA11 / Lutetium-177 (^{177}Lu) PSMA617 Theranostics

^{68}Ga-PSMA-PET using small molecules is a highly sensitive and specific test for the detection of prostate cancer. The most widely utilised agent, ^{68}Ga-PSMA11, appears superior to conventional

imaging and choline-PET in the staging and restaging of prostate cancer (52-54). Demonstration of high uptake on ^{68}Ga-PSMA-PET is used for selection of patients who are potentially suitable for ^{177}Lu-PSMA therapy. The diagnostic and therapeutic radiotracers are similar but the hydroxybenzyl ethylenediamine (HBED) chelator used to label ^{68}Ga to PSMA11 is not suitable for ^{177}Lu labelling. ^{177}Lu is labelled to PSMA617 using the DOTA chelator. The final product ^{177}Lu-DOTA-PSMA617 has lower renal uptake and high tumour internalisation compared to ^{68}Ga-HBED-PSMA11.

The reports in the literature of ^{177}Lu-PSMA617 are very encouraging despite their variable use in terms of both administered activity and the number of cycles administered. All of the studies included patients who were heavily pre-treated with 44-100% receiving prior systemic chemotherapy and 43-100% either abiraterone or enzalutamide. A significant number of patients had also received therapy with ^{223}Ra. Biochemical response (PSA reduction of greater than 50%) was observed in 31-72% of patients with progressive disease seen in 11-32 %. RECIST partial response are only reported in a few studies and varied from 11-40% with stable disease seen in 52-56%. Reported biochemical progression-free survival ranges from 126 to 175 days and overall survival has been reported ranging from 29 weeks to 15 months.

Expected toxicities following ^{177}Lu PSMA-617 relate to radiation damage to normal tissues that have PSMA expression or high delivery of radiation to sites of tumour with cross-fire effects to adjacent tissues. These toxicities include salivary and lacrimal gland dysfunction, nausea, flare effects at sites of tumours and bone marrow suppression. Haematological toxicity (grade 2-3) for haemoglobin was observed in 6.5-32% of patients, WBC 3-15% and platelets 0-25%. Other significant toxicity was limited to grade 1 xerostomia in 0-25%, grade 1 fatigue in 13-25%, and grade I nausea in 1.5-20%. A flare of symptoms such as pain owing to necrosis and consequent inflammation at sites of tumours may also occur. No acute renal toxicity has been reported although follow up is too short to determine rates of delayed renal toxicity.

No immediate acute severe adverse effects from the intravenous administration of unlabelled DKFZ-PSMA-617 have been described to date. The total administered amount of PSMA-ligand is less than 250 microgram. Specifically, no acute anaphylaxis has been reported.

Dosimetry studies looking at different small molecule inhibitors of PSMA (36, 55-57) demonstrate the salivary, lacrimal glands and kidneys receive the highest absorbed doses. They also predict that multiple doses of therapy could be safely administered without exceeding conventional normal tissue dose constraints. Data from the first 10 patients performed at Peter MacCallum Cancer Centre (PMCC) using a voxelized dosimetry tool based on quantitative SPECT/CT (58) performed at 4, 24 and 96hrs demonstrates concordant results. A summary of studies is shown in Table 1.

Table 1: Summary of published radiation dosimetry reports for ^{177}Lu radiolabelled small molecule inhibitors of PSMA. Absorbed dose mGy/MBq.

	No of patients	**Kidney**	**Salivary**	**Lacrimal**	**Marrow**	**Liver**
Zechmann 2014	16	1.45	4.6	-	0.36	1.47
Kabaskal 2015	7	0.88	0.88	-	0.03	-
Delker et al 2016	10	0.61	0.61	-	0.12	0.11
Hohburg et al 2016	9	0.53	0.53	2.82	-	-
Violet et al 2016(59)	10	0.43	0.43	-	0.2	0.08

Using the data from Violet et al.(59), assuming ‘average’ bio-distribution and an administration of 4 x 8 GBq activity, the predicted renal and salivary gland absorbed doses would be within the reported normal tissue dose constraints for salivary tissues and kidneys respectively (16.6 and 13.8 Gy); normal tissue constraints used in for these organs in external beam are 25 and 15-18 Gy respectively (60, 61).

In this 30 patient pilot study we have demonstrated excellent tumour response rates with a PSA

response rate (>50% reduction) of 57% (66). Grade 3 or higher haematological toxicity has been limited to 10-20% and there have been no reported incidences of neutropenic fever. Radiation dosimetry predicts that repeated therapies are feasible without compromising normal tissue dose constraints derived from external beam radiotherapy.

1.7 Comparator arm (Cabazitaxel)

Cabazitaxel is the first chemotherapy agent to improve survival in patients with mCRPC with progressive disease after docetaxel-based treatment, resulting in a 30% reduction in the risk of death and an improved median overall survival compared with mitoxantrone (8).

Cabazitaxel is a tubulin-binding taxane drug with anti-tumour activity in docetaxel-resistant cancers. In pre-clinical studies is an active agent against a number of prostate cancer cell lines including those that are taxane-resistant (62). It improves survival of men with mCRPC in comparison to mitoxantrone and is a current standard of care for docetaxel-resistant prostate cancer.

In the TROPIC phase III study (8), the hazard ratio for overall survival was 0.7 (95% confidence intervals 0.59-0.73 p <0.0001). Median survival was 15.1 months (95% CI 14.1–16.3) in the cabazitaxel group and 12.7 months (11.6–13.7) in the mitoxantrone group. Median progression-free survival was 2.8 months (95% CI 2.4–3.0) in the cabazitaxel group and 1.4 months (1.4–1.7) in the mitoxantrone group (HR 0.74, 95% confidence intervals 0.64–0.86, p<0.0001). Reduction in PSA by >50% was observed in 17.8% (13.7-22.0) of patients receiving mitoxantrone and 39.2% (33.9-44.5) of patients receiving cabazitaxel (p = 0.0002).

The most common toxic effects of cabazitaxel are haematological. The reported grade 3 neutropenia rate in TROPIC was 82% with a febrile neutropenia rate of 8%; the most frequent cause of death in the cabazitaxel group was neutropenia and its clinical consequences. The most common non-haematological grade 3 toxicity is diarrhoea. Grade 3 peripheral neuropathy is uncommon but low grade neuropathy is seen in 14% of patients and peripheral oedema is seen in around 10% of patients. Generally cabazitaxel is well tolerated in this patient population. Data presented from the PROSELICA trial in 2016 showed non-inferiority of the 20 mg/m^2 dose compared to the standard 25 mg/m^2 starting dose used in the TROPIC trial in terms of overall survival (72). Progression-free survival was also similar between the two doses. The higher dose arm demonstrated a higher numerical PSA response rate in the higher dose arm but also significantly higher toxicity, in particular grade 3 or 4 toxicity. The 20 mg/m^2 dose is now considered an appropriate starting dose for men considering cabazitaxel in the mCRPC setting.

2 AIM AND OBJECTIVES

The aim of this study is to determine and compare the activity and safety of treatment with Lu-PSMA versus cabazitaxel therapy in mCRPC.

The primary objective is to determine effects on:

1. Prostate Specific Antigen (PSA) response rate (PSA reduction of ≥50% from baseline).

The secondary objectives are to determine effects on:

2. Pain response (≥2 point reduction in PPI without increase in analgesic score) rate
3. Objective tumour response (OTR = CR or PR, RECIST 1.1and PCWG3) rate
4. Progression-Free Survival (PFS: time to PSA progression, pain progression, radiographic progression, or death)
5. PSA PFS (PCWG3)
6. Pain PFS (PPI and analgesia score)
7. Radiographic PFS (PCWG3 criteria for bone lesions and RECIST 1.1 for soft tissue lesions)
8. Aspects of health related quality of life (HRQL)
9. Overall survival (OS, death from any cause)
10. Frequency and severity of adverse events (CTCAE v 4.03)

The tertiary objectives of this study are to determine:

11. Associations between ^{68}Ga-PSMA PET/CT, FDG PET/CT, baseline characteristics, and outcomes
12. Associations between clinical outcomes and possible prognostic and/or predictive biomarkers (tissue and circulating) including ctDNA
13. Treatment and outcomes of participants excluded on basis of low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET

3 DESIGN

Open label, randomised, stratified, 2-arm, multicentre, phase 2 clinical trial.

Participants will be randomised to Lu-PSMA or cabazitaxel in a 1:1 ratio. A minimisation approach will be used to minimise chance imbalances across the following stratification factors: study site; volume of disease (>20 versus ≤20 sites of disease measured on ^{68}Ga-PSMA PET/CT); and, prior treatment with abiraterone or enzalutamide (yes vs no).

4 STUDY POPULATION

Patients must meet all of the inclusion criteria and none of the exclusion criteria to be eligible for this trial. No exceptions will be made to these eligibility requirements at the time of registration. All enquiries about eligibility should be addressed by contacting the NHMRC CTC prior to registration.

4.1 Target Population

Men with mCRPC suitable for cabazitaxel chemotherapy.

4.2 Inclusion Criteria

1. Male aged 18 or older with metastatic adenocarcinoma of the prostate defined by:
   - Documented histopathology of prostate adenocarcinoma

   OR

   - Metastatic disease typical of prostate cancer (i.e. involving bone or pelvic lymph nodes or para-aortic lymph nodes)

2. Castration-resistant prostate cancer (defined as disease progressing despite castration by orchiectomy or ongoing Luteinizing Hormone-Releasing Hormone (LHRH) analog)
3. Progressive disease with rising PSA defined by PCWG3 criteria (sequence of 2 rising values above a baseline at a minimum of 1-week intervals) and PSA ≥ 20 ng/mL
4. Target or non-target lesions according to RECIST 1.1 (Appendix 1)
5. Prior treatment with docetaxel
6. Significant PSMA avidity on ^{68}Ga-PSMA PET/CT, defined as
   a minimum uptake of SUVmax 20 at a site of disease, and
   SUVmax > 10 at sites of measurable disease ≥10mm (unless subject to factors explaining a lower uptake, e.g. respiratory motion, reconstruction artefact)
7. ECOG Performance status 0 to 2 (Appendix 2)
8. Assessed by a medical oncologist as suitable for chemotherapy with cabazitaxel
9. Adequate renal function:
   - Cr Cl ≥ 40mL/min (Cockcroft-Gault formula, Appendix 3)
10. Adequate bone marrow function:
    - Platelets ≥ 100 x10^9/L
    - Hb ≥ 90g/L (no red blood cell transfusion in last 4 weeks)
    - Neutrophils > 1.5 x10^9/L
11. Adequate liver function:
    - Bilirubin < 1.5 x upper limit of normal (ULN) (or if bilirubin is between 1.5-2x ULN, must have a normal conjugated bilirubin)
    - AST or ALT ≤ 2.0 x ULN (or ≤ 5.0 x ULN in the presence of liver metastases)
12. Estimated life expectancy > 12 weeks
13. Study treatment both planned and able to start within 21 days of randomisation
14. Willing and able to comply with all study requirements, including all treatments (cabazitaxel or Lu-PSMA); and, the timing and nature of all required assessments
15. Signed, written informed consent

4.3 Exclusion Criteria

1. Prostate cancer with known significant sarcomatoid or spindle cell or neuroendocrine small cell components
2. Site(s) of disease that are FDG positive with minimal PSMA expression defined as FDG intensity > ^{68}Ga-PSMA activity OR ^{68}Ga-PSMA SUVmax < 10
3. Sjogren's syndrome
4. Prior treatment with cabazitaxel or Lu-PSMA
5. Contraindications to the use of corticosteroid treatment
6. Active malignancy other than prostate cancer

7. Concurrent illness, including severe infection that may jeopardise the ability of the participant to undergo the procedures outlined in this protocol with reasonable safety
8. Presence of any psychological, familial, sociological or geographical condition potentially hampering compliance with the study protocol and follow-up schedule, including alcohol dependence or drug abuse
9. Patients who are sexually active and not willing/able to use medically acceptable forms of barrier contraception

4.4 Study Enrolment

4.4.1 **Pre-Screening**

Written informed consent must be signed and dated by the participant, and signed and dated by the Investigator, prior to any study-specific screening investigations being performed.

4.4.2 **Registration for ^{68}Ga-PSMA PET/CT and FDG PET/CT**

For registration to this study, potential participants must meet all of the eligibility criteria except those relating to imaging with ^{68}Ga-PSMA PET/CT and FDG PET/CT (inclusion 6 and exclusion 2).

After participants are registered for the study, imaging with ^{68}Ga-PSMA PET/CT and FDG PET/CT will be performed. These procedures including manufacture of ^{68}Ga-PSMA according to the study manual. The images will be completed at participating sites, and sent electronically in DICOM format to a central repository. The images will be centrally reviewed within 3 working days to determine eligibility for randomisation. Reason(s) for ineligibility will be recorded.

Participants who have provided consent but do not meet all the eligibility criteria will be recorded as screen failures.

4.4.3 **Randomisation**

Participants must continue to meet all of the inclusion criteria and none of the exclusion criteria to be randomised in this study.

Participants will be randomly allocated (1:1) to treatment with either Lu-PSMA or cabazitaxel. Study treatment should be planned to start within 21 days after randomisation.

Randomisation will be done according to the instructions in the Study Manual.

Once the randomisation process has been completed, the participant will be assigned a treatment group. Confirmation of each randomisation will be provided to the site.

Individuals may only be randomised once in this trial.

5 TREATMENT PLAN

Lu-PSMA is the experimental intervention in this trial and cabazitaxel chemotherapy in combination with oral prednisolone is the standard comparator (control) intervention.

Participants in both groups continue treatment with a LHRHA (or surgical castration), as per standard of care. Androgen deprivation with LHRHA (or surgical castration) is to be given continuously throughout this trial as required background treatment.

The dosing schedule for each treatment group is described in Table 2.

Table 2: Study Treatments

Agent(s)	Dose	Route	Duration
Lu-PSMA	6 to 8.5 GBq every 6 weeks	IV	Maximum of 6 doses or until participant is no longer clinically benefiting, or experiences unmanageable toxicity
Cabazitaxel and Prednisolone	20 mg/m^2 every 21 days 10mg daily	IV Orally	Maximum of 10 cycles or until participant is no longer clinically benefiting, or experiences unmanageable toxicity

5.1 Study Treatment: Lu-PSMA

5.1.1 **Lu-PSMA Administered Activity**

The administered activity starts at 8.5 G Bq in cycle 1, and is reduced by 0.5 GBq per cycle if there were no dose-limiting toxicities requiring an additional dose reduction (see Table 3). See section 5.3 for dose reductions and delays.

Table 3: Administered activity of Lu-PSMA

Cycle Number	**Administered Activity (GBq)**
1	8.5
2	8.0
3	7.5
4	7.0
5	6.5
6	6.0

The decision to order Lu-177 is required at least 10 days before the planned administration for each cycle.

Lu-PSMA will be administered by slow intravenous injection. SPECT/CT imaging encompassing the neck, chest, abdomen and pelvis will be performed 24 hours (+/- 4 hours) after Lu-PSMA administration. See Nuclear Medicine Manual for Lu-PSMA manufacturing, quality control, and administration details.

5.1.2 Maximum number of treatments with Lu-PSMA

Treatment will be administered to a maximum of 6 cycles, at a minimum interval of every 6 weeks (42 days), subject to availability.

5.1.3 Suspension of Lu-PSMA following exceptional response

Treatment with Lu-PSMA may be suspended if participants exhibit an exceptional response to treatment.

This is defined on the 24 hour post treatment SPECT/CT as marked reduction in uptake at all sites of disease with minimally-avid or non PSMA-avid. If a local site deems that the scan criteria has been met, these images will be centrally reviewed to confirm eligibility for treatment suspension.

5.1.4 Re-treatment with Lu-PSMA after suspension for exceptional response

Participants who experience an exceptional response, then experience disease progression may be considered for re-treatment with Lu-PSMA if the trial is still open and if they meet all of the following criteria:

- Have received < 6 doses of Lu-PSMA
- Symptomatic disease from prostate cancer OR PSA progression (PCWG3) OR radiological progression

Repeat imaging with ^{68}Ga-PSMA PET/CT and FDG PET/CT must be performed prior to re-treatment to assess eligibility. These images will be centrally reviewed to determine eligibility for re-treatment. Patients with sites of FDG-positive PSMA-negative sites of disease will not be suitable for re-treatment.

Participants who are eligible for re-treatment after a previous exceptional response must have a CT CAP and radioisotope bone scan performed if not already performed in the 28 day period prior to re-treatment. This will provide the new baseline from which response to re-treatment will be assessed.

Participants who are eligible for re-treatment with Lu-PSMA will receive dosing according to the administered activity due for the next cycle (for example if the participant received 3 cycles of Lu-PSMA prior to stopping due to an exceptional response, the next administered activity will be according to cycle 4) (section 5.1.1). Participants will be subject to the same dose modifications and stopping criteria.

5.1.5 Study Treatment: Cabazitaxel and Prednisolone

Cabazitaxel will be given every 3 weeks at a dose of 20mg/m^2, to a maximum of 10 cycles. It will be administered as a 60 minute IV infusion. It is not to be administered as an IV push or bolus injection.

Prednisolone 10mg orally daily will be administered throughout treatment with cabazitaxel. Prednisolone is to be continued regardless of dose delays of cabazitaxel and should not be stopped suddenly.

5.2 Required Background Treatment

All participants are to receive ongoing background androgen deprivation therapy with a LHRHA or surgical castration, as per standard of care. The choice of the LHRHA or surgical castration is at the discretion of the treating clinician.

Administration of the LHRHA should be according to the product information guide. Options include but are not restricted to: goserelin, leuprorelin, triptorelin, or degarelix.

5.3 Dose modifications and delays for Lu-PSMA

5.3.1 Reduction of administered activity of Lu-PSMA for Adverse Events

The next planned dose of Lu-PSMA will be reduced by 20% for participants who have a dose limiting toxicity in the preceding cycle. Dose limiting toxicities are defined as those considered both attributable to Lu-PSMA and dose-related (frequency and/or severity are related to the dose administered).

Dose-limiting toxicities for Lu-PSMA in this trial are defined as any one the following:

- nadir platelet count <100 x 10^9/L
- nadir neutrophil count <1.0 x 10^9/L
- dry mouth of grade 2 (moderate symptoms; oral intake alterations [e.g., copious water, other lubricants, diet limited to purees and/or soft, moist foods]) or worse
- dry eyes of grade 2 (symptomatic; multiple agents indicated; limiting instrumental ADL) or worse
- other significant dose-related toxicities, i.e. adverse events of grade 3 or worse that are considered both attributable to Lu-PSMA and dose related. The need for dose reductions and delays for these 'other toxicities' should be first discussed with the study team.

The administered dose of Lu-PSMA in the cycles after a 20% dose reduction are further reduced by 0.5 GBq in each successive cycle. There are no re-escalations of Lu-PSMA in this trial. There are no additional dose reductions beyond those specified above for dose-limiting toxicities that occur in subsequent cycles. If in doubt, please contact the study team.

For example, a participant with a nadir platelet count of 80 x 10^9/L in cycle 3 would have an administered dose of 6.0 GBq in cycle 4 (0.80 x 7.5 GBq = 6.0 GBq), 5.5 G Bq in cycle 5, and 5.0 GBq in cycle 6.

5.3.2 Dose delays for Lu-PSMA.

If the platelet count is <150 x 10^9/L 3 weeks after Lu-PSMA administration, then the platelet count should be re-checked one week later. If the platelet count is declining and less than 100 x 10^9/L, the next cycle of Lu-PSMA therapy will be deferred and bloods will be checked weekly. The next dose of Lu-PSMA should only be ordered when the platelet count is both rising and above 75 x 10^9/L. Administration of the next cycle can proceed if pre-treatment bloods show platelets >75 x 10^9/L.

Treatment should generally be withheld during AEs of severity Grade 3-4 (according to CTCAE v4.03) with the exception of fatigue or lymphocytopaenia, and not restarted until the adverse event has resolved to Grade 0-2. If in doubt, please contact the study team.

For AEs related to solely laboratory tests, the tests should be performed at least weekly to assess for recovery. The next dose of lutetium should not be ordered until the laboratory AE have resolved. Blood transfusion to correct anaemia is allowed in this trial.

5.4 Dose Modifications for Cabazitaxel

Instructions for treatment delays and dose modifications for AEs are specified below. In general, treatment should be withheld during AEs of severity Grade 3-4 (according to CTCAE v4.03), and not restarted until the adverse event has resolved to Grade 0-1, at the investigator's discretion. Day 1 treatment may be delayed for a maximum of 3 weeks. If the adverse event has not resolved to Grade 0-1 after delaying day 1 treatment for 3 weeks, then study treatment should be discontinued.

Specified dose reductions apply to all subsequent doses of study drug. Dose re-escalations after reductions for adverse events are prohibited.

If a patient experiences several adverse events with differing recommendations, then the modification that results in the longest delay and lowest dose should be used.

5.4.1 **Dose Levels**

No more than two dose reductions of cabazitaxel are allowed for any patient.

Table 4: Cabazitaxel dose levels

Dose Level	**Cabazitaxel (mg/m^2)**
Level 0	20 mg/m^2
Level -1	15 mg/m^2
Level -2	10 mg/m^2

5.4.2 **Dose Modifications for Adverse Events**

a) Myelosuppression

Dose modifications are to be made based on the granulocyte and/or platelet count drawn prior to planned treatment (can be done up to 3 days prior to planned dose):

Table 5: Cabazitaxel dose modifications for myelosuppression

Cabazitaxel	Neutrophils / 10^9/L Day 1 (within 7 days) of treatment		Platelets / 10^9/L Day 1(within+/- 7 days) of treatment
No change	≥ 1.5	and	≥ 100
Delay and reduce one dose level*	<1.5	or	<100

* If a dose is held due to myelosuppression, the patient will be retreated with a one level dose reduction once neutrophil count has recovered to at least 1.5 x 10^9/L and platelet count has recovered to at least 100 x 10^9/L.

Delay and dose modification after complicated neutropenia. Patients with afebrile Grade 4 neutropenia ≥ 7 days, or Grade 3-4 neutropenia associated with fever (one reading of oral temperature > 38.5°C, or three readings of oral temperature >38.0°C in a 24-hour period) can be retreated with a 1-level dose reduction once the absolute neutrophil count has increased to ≥ 1.5 x 10^9/L. The fever must have resolved and if an infection is identified, it must be adequately treated and have clinically resolved before restarting therapy. If prior bacteremia, blood cultures must be negative on recheck. Patient can continue with chemotherapy dosing while on antibiotics. Use of growth factors is not required as the dose and schedule does not meet ASCO guidelines. If however, the investigator considers it in the patient's best interest growth factors can be used per investigator discretion.

b) Hepatic dysfunction

ALT and Bilirubin will be evaluated pre-study and prior to each cycle (up to 7 days prior to planned dose) of cabazitaxel:

Patients who develop abnormal liver function tests for any reason while on the study will have the following dose reductions:

Table 6: Cabazitaxel Dose Modifications for Abnormal Liver Function

Bilirubin		ALT/ SGPT	Action
> ULN*	or	> 5 x ULN	Delay treatment (up to 3 weeks). If recovered**, reduce cabazitaxel dose by one dose level. If not, discontinue cabazitaxel.
≤ ULN*	and	> 3 x ULN	Reduce cabazitaxel by one dose level

* For patients with Gilbert's Syndrome, withhold dosing if the bilirubin level is >1.5 its baseline value

** Recovery is < 3X ULN for ALT/SGPT and within normal limits for bilirubin. For patients with Gilbert's Syndrome, recovery is defined as a bilirubin level <1.5 its baseline value. Dose modifications are based on ALT/ SGPT alone due to the lack of specificity of AST/SGOT.

c) Stomatitis

If stomatitis ≥ grade 2 is present on day 1 of any cycle, cabazitaxel should be held until stomatitis has resolved. If Grade 3/4 stomatitis occurs at any time, the dose of cabazitaxel will be reduced one dose level for all subsequent doses. If a second Grade 3/4 stomatitis event is incurred, cabazitaxel will be reduced one more dose level. If a third Grade 3/4 stomatitis event occurs, the cabazitaxel should be ceased.

d) Peripheral neuropathy

If ≥ Grade 3, the patient should discontinue cabazitaxel.

If Grade 2, the cabazitaxel should be held and the patient should be retreated upon recovery to a ≤ Grade 1 toxicity with a dose reduction of cabazitaxel by one level.

If Grade 2 or greater neurotoxicity persists for more than 3 weeks, the patient should discontinue cabazitaxel.

e) Hypersensitivity reactions for cabazitaxel

Cabazitaxel should be discontinued for Grade 4 hypersensitivity reactions. There are no dose reductions for cabazitaxel hypersensitivity reactions.

Grade 4 Hypersensitivity is defined as a reaction that is life threatening and requires pressor and/or ventilator support or shock associated with acidemia and impairing vital organ function due to tissue hypoperfusion.

Patients with two episodes of Grade 3 hypersensitivity reactions or one Grade 4 hypersensitivity reaction should discontinue cabazitaxel.

f) Diarrhoea

If patients experience >grade 2 diarrhoea and concurrent Grade 3 or 4 neutropenia, hold cabazitaxel until ANC>1000/mm^3 and diarrhoea ≤ Grade 2.

If patients experience significant diarrhoea (>3 loose stools/24hrs over baseline), they should be treated prophylactically in subsequent cycles with loperamide or diphenoxylate. If patient experiences > Grade 2 diarrhoea despite prophylaxis, cabazitaxel should be reduced one dose level. If patients experience > Grade 2 diarrhoea despite prophylaxis AND dose reduction, they should discontinue cabazitaxel.

g) Other toxic effects possibly related to cabazitaxel:

If toxicities ≤ Grade 2, manage the patient symptomatically if possible, and retreat without dose reduction.

If toxicities ≥ Grade 3 and clinically significant (not mentioned above), cabazitaxel should be withheld (except for anemia as patients can be transfused) until resolution to ≤ Grade 1 or baseline and patients treated with a one dose level reduction.

5.5 Post-study treatment

Treatment after completion or discontinuation of study treatment is at the discretion of the patient's treating clinicians.

Clinicians may choose to re-treat with cabazitaxel chemotherapy regardless of number of cabazitaxel cycles received.

5.6 Concomitant Medications

5.6.1 **Recommended**

The following medications and treatments are recommended in this study.

5.6.1.1 Recommended medications for use with Lu-PSMA

- Dexamethasone 8mg orally on day of Lu-PSMA injection at least 15 minutes prior to Lu-PSMA, and 4mg on day 2 and 3, for each Lu-PSMA cycle.
- Ondansetron (or equivalent) is recommended from day 1 to 3.
- Other anti-emetics should be used as required as per standard clinical practice.

5.6.1.2 Recommended medications for use with Cabazitaxel

- Anti-emetics should be used as required as per standard clinical practice.

  One recommended regimen is as follows: Oral premedication with Loratadine 10 mg (or other antihistamine) 60 minutes prior to cabazitaxel AND Ranitidine 150 mg (or other H2 antagonist) 60 minutes prior to cabazitaxel AND Dexamethasone 8 mg (or equivalent steroid) at least 60 minutes prior to cabazitaxel.

5.6.2 **Permitted**

The following medications and treatments are permitted in this study:

- Calcium and/or vitamin D supplementation.
- Treatment for bone metastases as per clinical guidelines, if commenced prior to randomisation and on a stable dose: zoledronic acid or other bisphosphonates, denosumab or other RANK-ligand inhibitors.
- Palliative radiotherapy is permitted without the requirement for withdrawal from study treatment, as long as the investigator deems the participant continues to derive clinical benefit from continued study treatment.

5.6.3 **Prohibited**

The following medications should not be used during this study. Participants who require treatment with any of these agents will usually need to discontinue study treatment. This should be discussed with the Study Chair by contacting the NHMRC CTC:

- Other investigational treatments
- Other anti-cancer therapy, including unproven treatments

- Over the counter therapies such as phytoestrogens

5.6.4 Concomitant Medication Reporting

Concomitant medications will not be recorded, except those administered within 30 days prior to a serious adverse event. Concomitant medications used to treat an SAE will not be recorded.

5.7 Compliance

Participant medication compliance will be determined by their attendance for study treatment administration.

5.8 Study Treatment Discontinuation

Study treatment will be permanently discontinued for any of the following reasons:

- Unacceptable toxicity as determined by the patient or site investigator or as defined in section 5.3 or 5.4.
- Multidisciplinary consultation (as a minimum: medical oncology and nuclear medicine physician) determines that continuation of treatment is not in the patient's best interest.
- Occurrence of an exclusion criterion affecting patient safety.
- Required use of a prohibited treatment (section 5.6.3).
- Significant protocol non-compliance, e.g. repeatedly failing to attend scheduled assessments.
- The participant declines further study treatment, or withdraws their consent to participate in the study.
- Evidence that the participant is no longer clinically benefiting

In addition Lu-PSMA will be discontinued for:

- Unequivocal progression, where the treating investigator determines that continuing treatment is unlikely to be of benefit.
- Delay of day 1 treatment of any cycle for > 16 weeks from planned day of next cycle due to treatment-related adverse events. For delays > 16 weeks due to reasons other than treatment-related adverse events, please contact the coordinating centre to discuss treatment continuation.

In addition Cabazitaxel will be discontinued for:

- Progressive disease (PD) according to PCWG3 (see Appendix 4) or imaging (RECIST 1.1, see Appendix 1) or symptomatic deterioration. As per PCWG3, treatment decisions should not be based only on PSA progression recorded in the first 12 weeks of study treatment alone. (See Appendix 4 for more details on the PCWG3 criteria).
- Delay of day 1 treatment of any cycle for > 21 days from planned day of next cycle due to treatment-related adverse events. For delays > 21 days due to due to reasons other than treatment-related adverse events, please contact the coordinating centre to discuss treatment continuation.

The reasons for discontinuing treatment will be documented in the participant's medical record and (e)CRF.

Participants who stop study treatment prior to the time recommended in the protocol will be requested to continue follow-up visits according to the protocol.

6 ASSESSMENT PLAN

6.1 Schedule of Assessments

Stage	**Screening**	**On treatment**		**Off treatment**[1]		**At Progression**[8] **(both PSA and radiological)**	**Post progression follow-up**	**Ineligible patients: post screen failure on PSMA/FDG PET**
Timing	Within 28 days prior to Randomisation	All participants ±7 days	Lu-PSMA only ±3 days	30-42 days after the last dose of study treatment	Follow-up until radiological progression ±7 days	±7 days	12 weekly ±7 days	Every 24 weeks (±4 weeks)
Informed Consent	X							
^{68}Ga-PSMA-PET/CT and FDG PET/CT	After registration (all other eligibility criteria met)							
SPECT/CT and Nuclear Medicine Review			24 h (+/- 4 h) after administration of Lu-PSMA					
CT CAP and Radioisotope Bone Scan[2]	X[3]	Weeks 11, 23, 35, then 12 weekly until radiological progression			12 weekly until radiological progression	X		
Blood Tests[4]	X	3 weekly		X	4 weekly			
Translational Research bloods[5]		Week 1 (C1D1) & Week 13, i.e. C3D1 Lu-PSMA or C5D1 cabazitaxel				At first progression (PSA or radiological), before subsequent therapy.		
PSA	X	3 weekly[6]		X	4 weekly	X		
Medical Oncologist Assessment	X	3 weekly		X	4 weekly	X		

Stage	Screening	On treatment		Off treatment[1]		At Progression[8] (both PSA and radiological)	Post progression follow-up	Ineligible patients: post screen failure on PSMA/FDG PET
Timing	Within 28 days prior to randomisation	All participants ±7 days	Lu-PSMA only ±3 days	30-42 days after the last dose of study treatment	Follow-up until radiological progression ±7 days	±7 days	12 weekly ±7 days	Every 24 weeks (±4 weeks)
Ordering of next cycle of Lu-PSMA			From day 21 after Lu-PSMA after d/w Med. Onc.and Nucl. Med. Phys.					
AE[7]		3 weekly		X	12 weeks only			
Concomitant medications		Drugs used prior to SAE						
HRQL forms (PPI, EORTC QLQ-C30, PDF)		3 weekly		X	4 weekly	X		
Survival and Subsequent Treatment							X[9]	X[9]

Notes

1. Either finished planned course of treatment, or stopped early due to exceptional response or toxicity.
2. Imaging must include a CT of the chest, abdomen and pelvis, and a radio-isotope whole body bone scan (WBBS).
3. If the baseline FDG PET imaging includes a diagnostic CT then there is no need to repeat a separate baseline CT scan.
4. Blood tests include the following
   Haematology: complete blood examination (CBE): Haemoglobin concentration, white cell count, platelet count, white cell differential.
   Biochemistry: electrolytes, urea, creatinine (EUC); liver function tests (LFT): bilirubin, alkaline phosphatase (ALP), alanine aminotransferase (ALT), gamma glutamyl transferase (GGT)
   If screening bloods are performed within 21 days of cycle 1, they are not required to be repeated prior to cycle 1.
5. Refer to Biological Sampling Manual for details of collection, processing and shipping procedures.
6. PSA must be repeated if performed more than 7 days before cycle 1.
7. Adverse events categorised and graded according to CTCAE v4.03 till 12 weeks after the study treatment ends.
8. At disease progression following an exceptional response, please refer to section 5.1.4 for details of the assessments required for participants eligible for re-treatment.
9. PSA response ≥ 50% will be recorded for the next-line of therapy administered post trial progression or ineligibility

6.2 Details of Assessments

6.2.1 **Clinical Assessment**

Clinical assessment includes history, physical examination, weight and performance status. Height should be measured at baseline. Patients will be reviewed by their study medical oncologist at least 3 weekly during their study treatment, and at least 4 weekly after study treatment until radiological progression.

6.2.2 **Imaging**

CT chest / abdomen / pelvis and ^{99m}Tc bone scan will be performed at screening, every twelve weeks while on treatment, and every twelve weeks after cessation of study treatment until radiological progression is demonstrated (unless radiological progression is documented prior).

6.2.3 **Quality of Life**

PPI, EORTC QLQ-C30 and PDF will be completed every 3 weeks until end of study treatment, and then every 4 weeks until radiological progression.

6.2.4 **Blood Collection**

Local pathology laboratories will be used for routine blood tests. Blood will be taken prior to the administration of treatment.

6.3 End of treatment and 30 day safety assessment

An end of treatment and safety assessment should be performed 30-42 days and 12 weeks after the last dose of study treatment to include any adverse events occurring within12 weeks after the last dose of study treatment. In the event of unresolved toxicity participants should continue to be followed.

6.4 Follow-up After Study Treatment Discontinuation

Study-specific follow-up assessments should be completed at the specified timepoints (± 7 days).

Participants who stop study treatment prior to the time recommended in the protocol will continue follow-up visits according to the protocol.

If a patient wishes to stop the study visits, they will be requested to allow their ongoing health status to be periodically reviewed via phone contact or from their general practitioner, or medical records.

6.5 Follow-up After Study Exclusion on Basis of low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET

Participants not randomised on the basis of low PSMA uptake or discordant FDG-avid disease, but that meet all other inclusion criteria and none of the exclusion criteria will have follow-up information collected via phone contact or medical records. Information collected will include survival status, the next line of anticancer treatment received and whether or not that treatment results in a PSA response of ≥50%.

7 OUTCOMES, ENDPOINTS AND OTHER MEASUREMENTS

7.1 PSA response rate (PSA RR)

The primary end point of this study is PSA RR defined as the proportion of participants in each group with a PSA reduction of ≥ 50% from baseline.

7.2 Pain Response (PPI and Analgesic Score)

Pain is measured using the McGill-Melzack Present Pain Intensity scale (PPI, see Appendix 5). Analgesic consumption will be converted to an analgesia score reflecting the oral Morphine Equivalent Daily Dose.

Pain Response is defined for patients with a baseline PPI score of ≥2 or a baseline analgesic score of ≥10 points, as: (i) a PPI score reduction of ≥2 points from baseline with no increase in analgesic score; and/or, (ii) a decrease of ≥ 50% in analgesic score with no increase PPI.

7.3 Objective tumour response (OTR = CR or PR, RECIST 1.1) rate

The objective tumour response rate (OTRR) is defined as the proportion of participants in each group with a confirmed complete response (CR) or partial response (PR) divided by the number of participants in each group, according to RECIST 1.1 (see Appendix 1).

7.4 Progression-Free Survival (time to PSA progression, pain progression, radiographic progression, or death)

Progression-free survival is defined as the interval from the date of randomisation to the date of first evidence of PSA progression, pain progression, radiographic progression, or death from any cause, whichever occurs first, or the date of last known follow-up without progression. The definition of each progression type is detailed in Sections 0 to 7.7.

Time to radiographic progression is defined as the interval from the date of randomisation to the date of first evidence of progression on imaging (PCWG3 criteria for bone lesions, see Appendix 4 and RECIST 1.1 for soft tissue lesions, see Appendix 1).

7.5 PSA progression-free survival (PCWG3)

PSA progression-free survival is defined as the interval from the date of randomisation to the date of first evidence of PSA progression or the date of last known follow-up without PSA progression.

PSA progression is defined as: a rise in PSA by more than 25% AND more than 2ng/mL above the nadir (lowest PSA point). This needs to be confirmed by a repeat PSA performed at least 3 weeks later.

Early rises in PSA prior to 12 weeks will be disregarded in determining PSA response (see Appendix 4 for more details on the PCWG3 criteria).

7.6 Pain progression-free survival (PPI scale and analgesia score)

Pain progression-free survival is defined as the interval from the date of randomisation to the date of first evidence of pain progression or the date of last known follow-up without pain progression.

Pain progression is defined as: (i) a ≥1 point increase in PPI score from nadir and (ii) a ≥25% an increase in analgesic score from nadir; or, (iii) the need for palliative radiotherapy.

7.7 Radiographic progression-free survival (PCWG3 / RECIST 1.1)

Radiographic progression-free survival is defined as the interval from the date of randomisation to the date of first evidence of radiographic progression or the date of last known follow-up without radiographic progression.

Time to radiographic progression is defined as the interval from the date of randomisation to the date of first evidence of progression on imaging (PCWG3 criteria for bone lesions and RECIST 1.1 for soft tissue lesions).

7.8 Aspects of Health related quality of life (HRQL)

Aspects of HRQL will be reported by participants using the EORTC core quality of life questionnaire (QLQ C-30, see Appendix 6), and the Patient Disease and Treatment Assessment Form (PDF, see appendix 7).

The QLQ-C30 is a validated questionnaire developed to assess HRQL in cancer patients. It includes five functional scales (physical, role, cognitive, emotional, and social), three symptom scales (fatigue, pain, and nausea and vomiting), and a global health and quality-of-life scale. The remaining single items assess additional symptoms commonly reported by cancer patients (dyspnoea, appetite loss, sleep disturbance, constipation, and diarrhoea), as well as the perceived financial impact of the disease and treatment.

Patient DATA Form (Patient Disease and Treatment Assessment Form – 47 items) a simple, multi-item quality of life instrument based on 11-point numeric rating scales for a range of relevant symptoms and functions. It combines cancer specific items from the UBQ-C, GLQ-8 and LASA cancer-specific quality of life instruments.[Stockler MR et al. Effect of sertraline on symptoms and survival in patients with advanced cancer, but without major depression: a placebo-controlled double-blind randomised trial. Lancet Oncol 2007; 8: 603-12].

7.9 Overall Survival (death from any cause)

Overall Survival is defined as the interval from the date of registration to date of death from any cause, or the date of last known follow-up alive.

7.10 Frequency and severity of adverse events (CTCAE v 4.03)

The CTCAE v4.03 will be used to classify and grade the intensity of adverse events during study treatment.

7.11 Associations between Ga-68 PSMA PET/CT, FDG-PET/CT baseline characteristics, and outcomes

Tests for associations of baseline quantitative parameters on Ga-68 PSMA PET/CT and FDG PET/CT will be performed. The parameters evaluated will include, but are not limited to SUVmax, PSMA-avid tumour volume defined by all disease with SUV greater than 3 for PSMA PET, metabolic tumour volume defined by all disease with SUV greater than liver SUVmean + 2 standard deviations and SUVmean of these tumour volumes. The prognostic value of the baseline parameters will be explored.

7.12 Associations between clinical outcomes and possible prognostic and/or predictive biomarkers (tissue and circulating) including ctDNA

Translational research studies will include identifying tissue and circulating biomarkers that are prognostic and/or predictive of response to treatment, safety and resistance to study treatment (associations of biomarkers with clinical outcomes). Studies may include, but are not limited to:

- ctDNA studies: analysing a mutation profile of key homologous recombination genes with PSA response and other clinical outcomes

7.13 Treatment and outcomes of participants excluded on basis of low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET

Overall Survival of this patient group will be recorded from the date of registration to date of death from any cause, or the date of last known follow-up alive.

8 SAFETY REPORTING

8.1 Definitions

An <u>ADVERSE EVENT</u> (AE) is any untoward medical occurrence in a patient or clinical investigational participant administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable or unintended sign (including an abnormal laboratory finding), symptom, or disease temporally

associated with the use of a medicinal investigational product, whether or not considered related to the medicinal product (see below).

AEs include the following:

- All suspected adverse drug or device reactions
- All reactions from drug or device – overdose, abuse, withdrawal, sensitivity, toxicity or failure of expected pharmacological action (if appropriate)
- Apparently unrelated illnesses
- Worsening (severity, frequency) of pre-existing illnesses or symptoms
- Injury or accidents
- Abnormalities in physiological testing or physical examination that require clinical intervention or further investigation (beyond ordering a repeat examination)
- Laboratory abnormalities that require clinical intervention or further investigation (beyond ordering a laboratory test)
- Any untoward event that occurs after the protocol-specified reporting period which the Investigator believes may be related to the drug or device.

A SERIOUS ADVERSE EVENT (SAE) is any untoward medical occurrence that at any dose:

- results in death,
- is life-threatening (i.e. the participant is at risk of death at the time of the event),
- requires inpatient hospitalisation or prolongation of existing hospitalisation,
- results in persistent or significant disability or incapacity,
- is a congenital anomaly/birth defect,
- other important medical events which, in the opinion of the investigator, are likely to become serious if untreated, or as defined in the protocol

NOTES:

(i) The term “life-threatening” in the definition of “serious” refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

(ii) Important medical events which may not be immediately life-threatening or result in death or hospitalization but which may jeopardize the patient or may require intervention to prevent one of the listed outcomes in the definition above should also be considered serious.

A SUSPECTED UNEXPECTED SERIOUS ADVERSE REACTION (SUSAR) is an SAE that is related to the drug or device and is unexpected, i.e. not listed in the investigator brochure or approved Product Information; or is not listed at the specificity or severity that has been observed; or is not consistent with the risk information described in the Participant Information Sheet and Informed Consent Form or elsewhere in the protocol. An event is causally related if there is a reasonable possibility that the drug [intervention] caused the AE, i.e. there is evidence to suggest a causal relationship between the drug and the event.

8.2 Recording Adverse Events

AEs will be recorded from the first dose of study treatment until 12 weeks after cessation of study treatment. The CTCAE v4.03 will be used to classify and grade the intensity of AEs after each treatment cycle. The worst grade will be recorded for each event. The investigator’s assessment of attribution to study drug administration will also be collected.

8.3 Reporting of Serious Adverse Events (including SUSARs)

The investigator is responsible for reporting all SAEs (including SUSARs) occurring during the study to the NHMRC CTC within 24 hours of investigational site staff becoming aware of the event according to the procedure documented in the Study Manual. SAEs must be reported up to 12 weeks from the end of study treatment.

The NHMRC CTC will be responsible for providing reports to the Lead HREC. The NHMRC CTC will provide SUSAR reports and SAE line listings to Investigators for submission to Human Research Ethics Committees (HRECs) as required. The investigator must notify the local HREC as required.

The NHMRC CTC will submit 'reportable safety events' to the TGA in Australia in time to comply with the requisite specified regulatory time windows (usually 7 days for fatal/life threatening events with an 8 day follow-up report, and 15 days for other SUSARs).

The following information will be recorded for each SAE**:**

- Event description including classification according to NCI CTCAE v4.03
- Reason for classification as an SAE (death, hospitalisation etc.)
- Severity / worst grade
- Attribution to study intervention (Lu-PSMA only)
- Expectedness (listed in IB/product information)
- Action taken with study intervention (Lu-PSMA or cabazitaxel), including rechallenge (if done)
- Outcome of SAE, including end date if recovered

8.4 Pregnancy

Pregnancy occurring in the partner of a participant and up to 90 days after the completion of the study treatment should also be reported to the investigator and the NHMRC CTC. The investigator should counsel the patient and their partner; discuss the risks of continuing with the pregnancy and the possible effects on the foetus.

The NHMRC CTC must be notified within 24 hours and the participant's partner followed during the entire course of the pregnancy and postpartum period. After obtaining participant and partner consent, parental and neonatal outcomes will be recorded even if they are completely normal.

9 CENTRAL REVIEW

9.1 Screening for eligibility

^{68}Ga-PSMA and FDG PET/CT images will be reviewed centrally to confirm the eligibility of participants and the suitability for re-treatment. SPECT/CT will be reviewed to confirm eligibility for suspension of Lu-PSMA treatment if participants exhibit an exceptional response. Details for submitting images are in the study manual.

PSMA 11 will be provided to each site to manufacture imaging of Ga-68-PSMA11.

9.2 Imaging

Reports and images of CT and bone scan imaging for tumour assessments will be submitted centrally. Details for submitting imaging reports are in the study manual.

10 NUCLEAR MEDICINE QUALITY ASSURANCE

Prior to beginning enrolment, all nuclear medicine imaging sites will be certified by the Australasian Radiopharmaceutical Trials Network (ARTnet). This will include certification of (1) ^{68}Ga-PSMA production, (2) ^{68}Ga-PSMA PET/CT phantom acquisition, (3) ^{177}Lu-PSMA617 production and (4) ^{177}Lu SPECT/CT phantom acquisition. Methods for these in addition to recommended radiation protection precautions are detailed in the study manual.

11 CENTRAL STORAGE OF BIOSPECIMENS

11.1 Central Blood Collection

Central laboratories will be used to conduct translational research studies including biomarker analyses. Translational research bloods will be collected and initially processed and stored at each site. Samples will later be shipped to a central facility for translational research and storage. Refer to the Biological Sampling Manual for procedures.

11.2 Central tissue collection

Archival tissue (from pre-study and any during study) may be collected for translational research from consenting participants. Samples will be shipped to central laboratories for translational research and storage. Refer to the Biological Sampling Manual for procedures.

12 STUDY TREATMENT SUPPLY AND ACCOUNTABILITY

12.1 Study Treatment

The investigational agent DKFZ-PSMA-617 is a small molecule ligand that binds to the extra-cellular domain of the prostate-specific membrane antigen. Each vial is supplied with a chemical purity ≥ 95%. Lutetium-177 (^{177}Lu) is a nuclear reactor-produced radiometal with a half-life of 6.7 days. Its decay mode is beta Max 0.5 MeV, average beta 0.13 MeV and gamma 208 keV (11% abundance). It is supplied in the chemical form of Lutetium Chloride ($LuCl_3$) in 0.04M HCl with a radionuclide and radiochemical purity of >99%.

No carrier added (n.c.a.) ^{177}Lu and DKFZ-PSMA-617 will be provided as clinical trial stock.

The final product ^{177}Lu-DKFZ-PSMA-617 will be compounded on site according to the Nuclear Medicine Manual by a qualified radiopharmacist or radiochemist. Quality control will be performed with HPLC prior to release of product. The study drug will be administered to eligible participants under the supervision of the nuclear medicine investigator or identified co-investigators.

12.2 Control Treatment

Cabazitaxel by Sanofi-Aventis is TGA-approved for this indication. Cabazitaxel will be provided to study participants according to usual hospital practice, and in Australia will be funded through the Pharmaceutical Benefits Scheme (PBS). Cabazitaxel will be dispensed as per standard pharmacy practice.

An in-line filter of 0.22 micrometer nominal pore size should be used during administration. Polyurethane infusion sets (tubing, filter, pumps) for the administration of the infusion solution should not be used.

Prednisolone will provided according to usual practice.

12.3 Drug Accountability

The radiopharmacy within the nuclear medicine department at participating institutions will maintain a record of Lutetium-177 and PSMA supply and use. A record will be maintained of the Lu-PSMA prepared for each patient. The radiopharmacy will also maintain a record of receipt and product destruction as appropriate.

Drug accountability does not need to be performed for cabazitaxel or prednisolone.

12.4 Background treatment

Androgen deprivation therapy will be provided according to usual practice. Drug accountability will not be performed for androgen deprivation therapy.

13 STATISTICAL CONSIDERATIONS

13.1 Sample Size

A trial of 200 participants randomised in 1:1 ratio to Lu-PSMA or cabazitaxel would have 80% power at the two-sided 5% level of significance to reject the null hypothesis of no difference between groups in PSA response rates if the true rates were actually 40% for cabazitaxel versus 60% for Lu-PSMA. The sample size incorporates an allowance of 3% for ineligible and/or inevaluable participants.

The study has 80% power at the two-sided 5% level of significance to reject the null hypothesis of no treatment effect on PFS if the true hazard ratio (HR) was 0.65, and assuming a median progression progression-free survival of 3 months in those allocated cabazitaxel, 24 months for accrual, and an additional 6 months for follow-up.

The study has 80% power at the two-sided 5% level of significance to reject the null hypothesis of no treatment effect on OS after 170 deaths have occurred if the true HR was 0.65. This number of deaths would be expected with 40 months of follow-up beyond the 24 months accrual period, assuming a median OS of 15 months in those allocated cabazitaxel and a true HR for OS of 0.65.

The estimates of PSA response, median PFS, and median OS for cabazitaxel are informed by the results of the TROPIC study (11). The hypothesised magnitude of the additional benefit to these endpoints with Lu-PSMA are judged to be clinically plausible and worthwhile.

13.2 Statistical Analysis

A statistical analysis plan will be finalised prior to data-lock, and contain additional detail on the methods described below.

All randomised participants will be eligible for inclusion in the full analysis set. Analysis of efficacy endpoints will be undertaken on participants in the full analysis set. A sensitivity analysis using a per-protocol analysis set may be performed on efficacy endpoints. The safety population will comprise all randomised participants who received at least one administration of study medication. Participants will be analysed according to the regimen they actually received for the purposes of the safety analysis.

All p-values and confidence intervals will be two-sided.

The primary analysis will be a comparison of the treatment groups on PSA response rate using a Cochran–Mantel–Haenszel $\chi 2$ test accounting for the stratification factors used at randomisation. Multivariable logistic regression will be used to estimate odds ratios with confidence internals for covariates and assigned treatment. Other binary endpoints (e.g. pain response) will be analysed in the same way.

Point estimates for time-to-event endpoints will be estimated using the Kaplan-Meier method with appropriate confidence intervals. Kaplan-Meier curves will be produced to summarise the

distribution of the time-to-event data. A log-rank test accounting for stratification factors will be used to compare time-to-event endpoints between groups. Cox Proportional Hazards regression will be used to estimate hazard ratios with confidence intervals.

Scale scores from the QLQ-C30 and PDF will be summarised by group over time. The applicability of repeated measures mixed modelling to these scales will be investigated. This will involve fitting a linear mixed model with patient as the random effect, and fixed effect terms for treatment allocation, time point, a time point-by-treatment allocation interaction, and the baseline assessment.

The relevant linear modelling approach will be used to address tertiary objectives, and identify prognostic and predictive factors. These tertiary analysis will include a comparison, adjusted for potential confounders, between patients randomised to cabazitaxel versus patients with low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET who are not randomised but go on to receive cabazitaxel.

The analysis of safety data will be principally descriptive in nature. Adverse Events (AEs) and Serious Adverse Events (SAEs) will be tabulated by treatment allocation and CTCAE criteria including system organ class, term, and worst grade.

13.3 Interim analyses

No interim analyses of efficacy parameters are currently planned but may be requested by the Trial Management Committee (TMC) if judged to be in patients' best interest given new/unanticipated circumstances arising. Such analyses would be overseen and interpreted by the ANZUP Independent Data Safety Monitoring Committee (IDSMC) using appropriate methods documented in the IDSMC charter.

14 STUDY ORGANISATION and COMMITTEES

14.1 Study coordination

The study is a locally developed and led investigator initiated collaborative group study. Coordination, monitoring, data acquisition and management and statistical analysis will be performed by the NHMRC CTC.

14.2 Trial Management Committee

The Trial Management Committee (TMC) will oversee study planning, monitoring, progress, review of information from related research, and implementation of recommendations from other study committees and external bodies (e.g. ethics committees).

14.3 Independent Data Safety Monitoring Committee

The ANZUP Independent Data Safety Monitoring Committee (IDSMC) will have independent oversight of the trial and monitor accrual, event rates, and key safety endpoints in accordance with the charter.

15 ADMINISTRATIVE ASPECTS

15.1 Ethics and regulatory compliance

This study will be conducted according to the Note for Guidance on Good Clinical Practice (CPMP/ICH/135/95) annotated with TGA comments (Therapeutic Goods Administration DSEB July 2000) and in compliance with applicable laws and regulations including the Code of Practice for the Exposure of Humans to Ionizing Radiation for Research Purposes (2005) (link

https://www.arpansa.gov.au/regulation-and-licensing/regulatory-publications/radiation-protection-series/codes-and-standards/rps8).

The study will be performed in accordance with the NHMRC National Statement on Ethical Conduct in Human Research 2007, the NHMRC Australian Code for the Responsible Conduct of Research 2007, and the principles laid down by the World Medical Assembly in the Declaration of Helsinki 2008.

To this end, no patient will be recruited to the study until all the necessary approvals have been obtained and the patient has provided written informed consent. Further, the investigator shall comply with the protocol, except when a protocol deviation is required to eliminate immediate hazard to a participant. In this circumstance the NHMRC CTC, principal investigator and HREC must be advised immediately.

15.2 Confidentiality

The study will be conducted in accordance with applicable Privacy Acts and Regulations. All data generated in this study will remain confidential. All information will be stored securely at the NHMRC CTC, University of Sydney and will only be available to people directly involved with the study and who have signed a Confidentiality Agreement.

15.3 Protocol amendments

Changes and amendments to the protocol can only be made by the Trial Management Committee. Approval of amendments by the Institutional HREC is required prior to their implementation. In some instances, an amendment may require a change to a consent form. The Investigator must receive approval/advice of the revised consent form prior to implementation of the change. In addition, changes to the data collected, if required, will be incorporated in the amendment.

The investigator should not implement any changes to, or deviations from, the protocol except where necessary to eliminate immediate hazard(s) to trial participant(s).

15.4 Data Handling and Record Keeping

All trial data required for the monitoring and analysis of the study will be recorded on the (e)CRFs provided. All required data entry fields must be completed. Data corrections will be done according to the instructions provided. The investigator will be asked to confirm the accuracy of completed CRFs by signing key CRFs as indicated.

Source documents pertaining to the trial must be maintained by investigational sites. Source documents may include a participant's medical records, hospital charts, clinic charts, the investigator's participant study files, as well as the results of diagnostic tests such as X-rays, laboratory tests, and electrocardiograms. The investigator's copy of the case report forms serves as part of the investigator's record of a participant's study-related data.

The following information should be entered into the participant's medical record:

a. The participant’s protocol identification.

b. The date that the participant entered the study, and participant number.

c. A statement that informed consent was obtained (including the date)

d. Relevant medical history

e. Dates of all participant visits and results of key trial parameters.

f. Occurrence and status of any adverse events

g. The date the participant exited the study, and a notation as to whether the participant completed the study or reason for discontinuation.

All study-related documentation at Australian sites will be maintained for 15 years following completion of the study.

15.5 Study Monitoring

Data from this study will be monitored by Clinical Trials Program staff from the NHMRC CTC or their delegates. Monitoring will include centralised review of CRFs and other study documents for protocol compliance, data accuracy and completeness. Monitoring may include monitoring visits to investigational sites during the study for source data verification, review of the investigator's site file and drug handling records. The NHMRC CTC will be given direct access to source documents, CRFs and other study-related documents. By signing the informed consent form, the participant gives authorised NHMRC CTC staff or their delegates direct access to their medical records and the study data.

15.6 Audit and Inspection

This study may be participant to audit or inspection by representatives of the Sponsor, the NHMRC CTC or representatives of regulatory bodies (e.g. Therapeutic Goods Administration (TGA).

15.7 Clinical Study Report

A Clinical Study Report which summarises and interprets all the pertinent study data collected will be issued which may form the basis of a manuscript intended for publication. The Clinical Study Report or summary thereof will be provided to the study investigators and the ethics committees. A lay summary of results will be prepared for patients and other interested parties.

15.8 Publication Policy

The ANZUP Publication Policy will apply. The Trial Management Committee will appoint a Writing Committee to draft manuscript(s) based on the trial data. Manuscript(s) will be submitted to peer-reviewed journal(s). The first publication will be the report of the full trial results based on the main protocol using the study group name, with subsequent publications of data subsets in individual names based on contribution. The Writing Committee will develop a publication plan, including authorship, target journals and expected dates of publication. All publications must receive prior written approval from the TMC prior to submission.

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17 APPENDIX

17.1 Appendix 1: Response Evaluation Criteria in Solid Tumours (RECIST 1.1)

These instructions are based on the guidelines recommended by Eisenhauer et al.(63)
The sections that apply to this trial are the criteria for progression of soft tissue lesions.

1 Evaluable for response.
All patients who have received at least one cycle of therapy and have their disease re-evaluated will be considered evaluable for response (exceptions will be those who exhibit objective disease progression prior to the end of cycle 1 who will also be considered evaluable). Patients on therapy for at least this period and who meet the other listed criteria will have their response classified according to the definitions set out below

2 Disease and lesion definitions
1.1 *Measurable Disease*. Measurable *tumour lesions* are defined as those that can be accurately measured in at least one dimension (longest diameter to be recorded) as ≥ 20 mm with chest x-ray, and as ≥10 mm with CT scan (assuming slice thickness of 5mm or less) or clinical examination. Bone lesions are considered measurable only if assessed by CT scan and have an identifiable soft tissue component that meets these requirements (soft tissue component > 10 mm by CT scan). *Malignant lymph nodes* must be ≥ 15mm in the short axis to be considered measurable; only the short axis will be measured and followed. All tumour measurements must be recorded in millimetres. Previously irradiated lesions are not considered measurable unless progression has been documented in the lesion.

1.2 *Non-measurable Disease*. All other lesions (or sites of disease), including small lesions are considered non-measurable disease. Bone lesions without a measurable soft tissue component, leptomeningeal disease, ascites, pleural/pericardial effusions, lymphangitis cutis/pulmonis, inflammatory breast disease, lymphangitic involvement of lung or skin and abdominal masses followed by clinical examination are all non-measurable. Lesions in previously irradiated areas are non-measurable, unless progression has been demonstrated.

1.3 *Target Lesions*. When more than one measurable tumour lesion is present at baseline all lesions up to *a maximum of 5 lesions in total* (and a maximum of *2 lesions per organ*) representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline. Target lesions should be selected on the basis of their size (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to *reproducible repeated measurements*. Note that pathological lymph nodes must meet the criterion of having a short axis of ≥ 15 mm by CT scan and only the *short* axis of these lymph nodes will contribute to the baseline sum. All other pathological lymph nodes (those with a short axis ≥ 10 mm but <15 mm) should be considered non-target lesions. Nodes that have a short axis < 10 mm are considered non-pathological and should not be recorded or followed (see 10.2.4). At baseline, the sum of the target lesions (longest diameter of tumour lesions plus short axis of target lymph nodes: overall maximum of 5) is to be recorded.

After baseline, a value should be provided on the CRF for all identified target lesions for each assessment, even if very small. If extremely small and faint lesions can not be accurately measured but are deemed to be present, a default value of 5 mm may be used. If lesions are too small to measure and indeed are believed to be absent, a default value of 0 mm may be used.
1.4 *Non-target Lesions*. All non-measurable lesions (or sites of disease) plus any measurable lesions over and above those listed as target lesions are considered *non-target lesions*. Measurements are not required but these lesions should be noted at baseline and should be followed as “present” or “absent”.

Response Definitions

All patients will have their BEST RESPONSE from the start of study treatment until the end of treatment classified as outlined below:
Complete Response (CR): disappearance of all *target* and *non-target* lesions and normalization of any specified tumour markers (no tumour markers for this trial). Pathological lymph nodes must have short axis measures < 10mm (Note: continue to record the measurement even if < 10mm and considered CR). Residual lesions (other than nodes < 10mm) thought to be non-malignant should be further investigated (by cytology or PET scans) before CR can be accepted. Confirmation of response is sometimes required in studies where objective tumour response is the primary endpoint, and the details of confirmation are then specified in the body of the protocol. Confirmation of response is not required in this study.

Partial Response (PR): at least a 30% decrease in the sum of measures for target lesions (longest diameter for tumour lesions and short axis measure for target lymph nodes), taking as reference the baseline sum of diameters. Non target lesions must be non-PD. Confirmation of response is sometimes required in studies where objective tumour response is the primary endpoint, and the details of confirmation are then specified in the body of the protocol. Confirmation of response is not required in this study

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD taking as reference the smallest sum of diameters on study.

Progressive Disease (PD): at least a 20% increase in the sum of diameters of measured lesions taking as references the smallest sum of diameters recorded on study (including baseline) AND an absolute increase of ≥ 5mm. Appearance of new lesions will also constitute progressive disease (including lesions in previously unassessed areas). In exceptional circumstances, unequivocal progression of non-target disease may be accepted as evidence of disease progression, where the overall tumour burden has increased sufficiently to merit discontinuation of treatment or where the tumour burden appears to have increased by at least 73% in volume. Modest increases in the size of one or more non-target lesions are NOT considered unequivocal progression. If the evidence of PD is equivocal (target or non-target), treatment may continue until the next assessment, but if confirmed, the earlier date must be used.

Table: Integration of Target, non-Target and New lesions into response assessment:

Target Lesions	Non-Target Lesions	New Lesions	Overall Response	Best Response for this category also requires
Target lesions ± non target lesions				
CR	CR	No	CR	Normalization of specified tumour markers, AND lymph nodes <10mm
CR	Non-CR/Non-PD	No	PR	
CR	Not all evaluated	No	PR	
PR	Non-PD/ not all evaluated	No	PR	
SD	Non-PD/ not all evaluated	No	SD	documented at least once ≥ 4 wks. from baseline [note, protocol may define; 6-8 weeks is recommended]
Not all evaluated	Non-PD	No	NE	
PD	Any	Any	PD	
Any	PD	Any	PD	
Any	Any	Yes	PD	
Non target lesions ONLY				
No Target	CR	No	CR	Normalization of specified tumour markers AND lymph nodes < 10mm
No Target	Non-CR/non-PD	No	Non-CR/non-PD	

No Target	Not all evaluated	No	NE	
No Target	Unequivocal PD	Any	PD	
No Target	Any	Yes	PD	
Note: Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as "*symptomatic deterioration*". This is a reason for stopping therapy, but is NOT objective PD. Every effort should be made to document the objective progression even after discontinuation of treatment.				

1 Response Duration

Response duration will be measured from the time measurement criteria for CR/PR (whichever is first recorded) are first met until the first date that recurrent or progressive disease is objectively documented, taking as reference the smallest measurements recorded on study (including baseline).

2 Stable Disease Duration

Stable disease duration will be measured from the time of start of treatment (or randomisation for randomized studies) until the criteria for progression are met, taking as reference the smallest sum on study (including baseline).

3 Methods of Measurement

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Assessments should be identified on a calendar schedule and should not be affected by delays in therapy, which may be treatment arm dependent, unless the protocol specifies otherwise. While on study, all lesions recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (e.g. 2 mm). If it is the opinion of the radiologist that the lesion has likely disappeared, the measurement should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. For lesions which fragment/split add together the longest diameters of the fragmented portions; for lesions which coalesce, measure the maximal longest diameter for the "merged lesion".

3.1 *Clinical Lesions*. Clinical lesions will only be considered measurable when they are superficial and ≥ 10mm as assessed using calipers (e.g. skin nodules). For the case of skin lesions, documentation by colour photography including a ruler to estimate the size of the lesion is recommended. If feasible, imaging is preferred.

3.2 *Chest X-ray*. Chest CT is preferred over chest X-ray, particularly when progression is an important endpoint, since CT is more sensitive than X-ray, particularly in identifying new lesions. However, lesions ≥ 20 mm on chest X-ray may be considered measurable if they are clearly defined and surrounded by aerated lung.

3.3 *CT, MRI*. CT is the best currently available and reproducible method to measure lesions selected for response assessment. This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. When CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g. for body scans). While PET scans are not considered adequate to measure lesions, PET-CT scans may be used providing that the measures are obtained from the CT scan and the CT scan is of identical diagnostic quality to a diagnostic CT (with IV and oral contrast).

3.4 *Ultrasound*. Ultrasound is not useful in assessment of lesion size and should not be used as a method of measurement. If new lesions are identified by ultrasound in the course of the study, confirmation by CT is advised.

3.5 *Endoscopy, Laparoscopy*. The utilization of these techniques for objective tumor evaluation is not advised. However, they can be useful to confirm complete pathological response when biopsies are obtained or to determine relapse in trials where recurrence following complete response or surgical resection is an endpoint.

3.6 *Tumour Markers*. Tumour markers alone cannot be used to assess objective tumor response. If markers are initially above the upper normal limit, however, they must normalize for a patient to be considered in complete response. There are no specified tumour markers for this trial.

3.7 *Cytology, Histology*. These techniques can be used to differentiate between PR and CR in rare cases if required by protocol (for example, residual lesions in tumour types such as germ cell tumours, where known residual benign tumours can remain). When effusions are known to be a potential adverse effect of treatment (e.g. with certain taxane compounds or angiogenesis inhibitors), the cytological confirmation of the neoplastic origin of any effusion that appears or worsens during treatment when the measurable tumor has met criteria for response or stable disease is advised to differentiate between response or stable disease and progressive disease.

17.2 Appendix 2: ECOG Performance Status

These scales and criteria are used by doctors and researchers to assess how a patient's disease is progressing, assess how the disease affects the daily living abilities of the patient, and determine appropriate treatment and prognosis. They are included here for health care professionals to access.

Grade	ECOG
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours
3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
5	Dead

* As published in Am. J. Clin. Oncol 1982(64).

17.3 Appendix 3: Cockroft-Gault formula

Renal function (GFR) may be estimated with the Cockcroft–Gault formula, as follows:

Male participants:

Creatinine clearance (ml/minute) = $\frac{(140 - age) * weight}{0.814 * SerumCr}$

Units:
Age in years
Weight in kilograms
Serum creatinine (SerumCr) in micromoles per litre

Female participants: Use above formula but multiply calculated Creatinine clearance by 0.85

17.4 Appendix 4: Prostate Cancer Working Group 3 (PCWG3) Criteria

The sections that apply to this trial are the criteria for PSA response and progression, and the criteria for bone lesion "prevent/delay end points (progression).

Variable	PCWG3 (2016)
PSA	- Recognize that a favorable effect on PSA may be delayed for 12 weeks or more, even for a cytotoxic drug - Monitor PSA by cycle but plan to continue through early rises for a minimum of 12 weeks unless other evidence of progression - Ignore early rises (prior to 12 weeks) in determining PSA response **For control/relieve/eliminate endpoints:** - Describe absolute changes in PSA over time from baseline to best response **For delay/prevent endpoints:** **Decline from baseline**: - Record time from start of therapy to first PSA increase that is ≥ 25% and ≥ 2 ng/mL above the nadir, and which is confirmed by a second value 3 or more weeks later (ie, a confirmed rising trend) **No decline from baseline:** - PSA progression ≥ 25% and ≥ 2 ng/mL after 12 weeks
Soft-tissue lesions	**For control/relieve/eliminate end points:** Use RECIST with caveats: - Record up to 5 lesions per site of disease - Record changes in nodal, lung, liver adrenal and CNS sites separately - Only report changes in lymph nodes that were ≥ 1.5 cm in diameter in short axis at baseline - Record changes in pelvic (regional) nodes vs extrapelvic (distant/metastatic) nodes separately - Only report changes in visceral lesions (liver, lung, adrenal, CNS) that were ≥ 1.0 cm in the longest dimension - Record complete elimination of disease at any site separately - Confirm favorable change with second scan - Record changes using waterfall plot **For delay/prevent end points:** - Record changes in nodal and visceral disease separately - Record up to 5 lesions per site of spread - Use RECIST 1.1 criteria for progression, but clearly record type of progression (growth of existing lesions vs development of new lesions) separately by site. with additional requirement that progression at first assessment be confirmed by a second scan 6 or more weeks later. (Particularly important when anticipated effect on PSA is delayed or for biologic therapies) - Previously normal (<1.0 cm) lymph nodes must have grown by ≥ 5 mm in the short axis from baseline or nadir and be ≥ 1.0 cm in the short axis to be considered to have progressed. Nodes that have progressed to 1.0 to less than 1.5 cm are pathologic, subject to clinical discretion, and non measurable. For existing pathologic adenopathy (≥ 1.5 cm), progression is defined per RECIST 1.1 - Note that for some treatments, a lesion may increase in size before it decreases. - Report the proportion who have not progressed at fixed time points (6 or 12 months)

Bone	**For control/relieve eliminate end points:**
	- Record outcome as new lesions, no new lesions or resolved lesion
	- First scheduled reassessment:
	  o No new lesions: continue therapy
	  o New lesions: perform a confirmatory scan 6 or more weeks later
	- Confirmatory scan:
	  o No new lesions: continue therapy
	  o Additional new lesions: progression
	- Subsequent scheduled reassessments:
	  o No new lesions: continue
	  o New lesions: progression
	- Changes in intensity or uptake do not constitute regression or progression
	For prevent/delay end points (progression):
	- The appearance of 2 or more new lesions, and, for the first reassessment only, a confirmatory scan performed 6 or more weeks later that shows a minimum of 2 or more additional new lesions
	- The date of progression is the date of the first scan that shows the change
	- Report the proportion of patients who have not progressed at fixed time intervals (6 and 12 months)

Symptoms	- Pain palliation assessment requires a patient population with clinically meaningful pain at baseline (eg, ≥ 4 on a 10-point pain intensity scale) and response defined as a clinically meaningful score improvement at a subsequent time point (eg, a 30% relative or 2-point absolute improvement from baseline at 12 weeks, confirmed at least 2 weeks later, without an overall increase in opiate use). **For control/relieve eliminate end points:** - Serial (eg, daily x 7 days) assessments at each time point can improve the stability of values Principles may be extended for any PRO for which a clinically meaningful baseline PRO score has been determined together with a responder definition that is based on a sustained clinically meaningful score improvement. **For delay/prevent end points:** Patients with any level of baseline pain, including no pain, are eligible to be evaluated for prevent/delay end points; those without pain are followed for development of pain, whereas those with baseline pain are followed for progression (eg, a 2-point increase without an overall decrease in opiate use). Pain assessment should be administered at treatment discontinuation and once again if feasible (eg, 2 to 4 weeks later). Time to deterioration of physical function and/or HRQoL scores should also be included, with a priori thresholds defining clinically meaningful deterioration score changes that are based on prior published data for the selected questionnaire.

See Scher et al 2016(65) for more details.

17.5 Appendix 5: McGill-Melzack Present Pain Intensity Scale (PPI)

1. Please circle one number to best describe your pain on average over the last 7 days.					
0	1	2	3	4	5
No pain	Mild	Discomforting	Distressing	Horrible	Excruciating

17.6 Appendix 6: EORTC QLQ C-30

ENGLISH

EORTC QLQ-C30 (version 3)

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

Please fill in your initials: _ _ _ _

Your birthdate (Day, Month, Year): _ _ _ _ _ _ _ _

Today's date (Day, Month, Year): 31 _ _ _ _ _ _ _ _

		Not at All	A Little	Quite a Bit	Very Much
1.	Do you have any trouble doing strenuous activities, like carrying a heavy shopping bag or a suitcase?	1	2	3	4
2.	Do you have any trouble taking a long walk?	1	2	3	4
3.	Do you have any trouble taking a short walk outside of the house?	1	2	3	4
4.	Do you need to stay in bed or a chair during the day?	1	2	3	4
5.	Do you need help with eating, dressing, washing yourself or using the toilet?	1	2	3	4

During the past week:

		Not at All	A Little	Quite a Bit	Very Much
6.	Were you limited in doing either your work or other daily activities?	1	2	3	4
7.	Were you limited in pursuing your hobbies or other leisure time activities?	1	2	3	4
8.	Were you short of breath?	1	2	3	4
9.	Have you had pain?	1	2	3	4
10.	Did you need to rest?	1	2	3	4
11.	Have you had trouble sleeping?	1	2	3	4
12.	Have you felt weak?	1	2	3	4
13.	Have you lacked appetite?	1	2	3	4
14.	Have you felt nauseated?	1	2	3	4
15.	Have you vomited?	1	2	3	4
16.	Have you been constipated?	1	2	3	4

Please go on to the next page

During the past week:	Not at All	A Little	Quite a Bit	Very Much
17. Have you had diarrhea?	1	2	3	4
18. Were you tired?	1	2	3	4
19. Did pain interfere with your daily activities?	1	2	3	4
20. Have you had difficulty in concentrating on things, like reading a newspaper or watching television?	1	2	3	4
21. Did you feel tense?	1	2	3	4
22. Did you worry?	1	2	3	4
23. Did you feel irritable?	1	2	3	4
24. Did you feel depressed?	1	2	3	4
25. Have you had difficulty remembering things?	1	2	3	4
26. Has your physical condition or medical treatment interfered with your family life?	1	2	3	4
27. Has your physical condition or medical treatment interfered with your social activities?	1	2	3	4
28. Has your physical condition or medical treatment caused you financial difficulties?	1	2	3	4

For the following questions please circle the number between 1 and 7 that best applies to you

29. How would you rate your overall health during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

30. How would you rate your overall quality of life during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

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17.7 *Appendix 7: Patient Disease and Treatment Assessment Form (PDF).*

Patient's Disease and Treatment Assessment Form – General

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Patient initials: ☐☐☐ (First middle last) MRN ☐☐☐☐☐☐☐ Today's date: ☐☐ (D D) ☐☐ (M M) ☐☐ (Y Y)

Please **circle** one number for each line to best show **how much that aspect troubled you** on average **during the last 3 weeks.**

	No trouble at all		Mild			Moderate			Severe		Worst I can imagine
1. Pain (all and anywhere)	0	1	2	3	4	5	6	7	8	9	10
2. Fatigue (tiredness)	0	1	2	3	4	5	6	7	8	9	10
3. Poor appetite	0	1	2	3	4	5	6	7	8	9	10
4. Cough	0	1	2	3	4	5	6	7	8	9	10
5. Shortness of breath	0	1	2	3	4	5	6	7	8	9	10
6. Trouble sleeping	0	1	2	3	4	5	6	7	8	9	10
7. Nausea	0	1	2	3	4	5	6	7	8	9	10
8. Vomiting	0	1	2	3	4	5	6	7	8	9	10
9. Diarrhoea	0	1	2	3	4	5	6	7	8	9	10
10. Constipation	0	1	2	3	4	5	6	7	8	9	10
11. Urinary symptoms	0	1	2	3	4	5	6	7	8	9	10
12. Leg swelling	0	1	2	3	4	5	6	7	8	9	10
13. Difficulty walking	0	1	2	3	4	5	6	7	8	9	10
14. Anxiety (feeling worried)	0	1	2	3	4	5	6	7	8	9	10
15. Depression (feeling sad)	0	1	2	3	4	5	6	7	8	9	10
16. Irritability (being cranky)	0	1	2	3	4	5	6	7	8	9	10
17. Trouble concentrating	0	1	2	3	4	5	6	7	8	9	10

Please circle one number for each line to show how you would rate yourself on that aspect on average **during the last month.**

	Best possible	Very Good		Good		Fair		Poor		Very poor	Worst possible
18. Energy	10	9	8	7	6	5	4	3	2	1	0
19. Appetite	10	9	8	7	6	5	4	3	2	1	0
20. Mobility (ability to get around)	10	9	8	7	6	5	4	3	2	1	0
21. Mood	10	9	8	7	6	5	4	3	2	1	0
22. Physical well-being	10	9	8	7	6	5	4	3	2	1	0
23. Emotional well-being	10	9	8	7	6	5	4	3	2	1	0
24. Overall well-being	10	9	8	7	6	5	4	3	2	1	0

Please continue, there are more questions on the next page.

Patient's Disease and Treatment Assessment Form – Specific

Patient initials ☐☐☐ (First middle last) MRN ☐☐☐☐☐☐☐ Today's date ☐☐ (D D) ☐☐ (M M) ☐☐ (Y Y)

Please circle one number for each line to best show **how much that aspect troubled you** on average **during the last 3 weeks.**

	No trouble at all		Mild			Moderate			Severe		Worst I can imagine
25. Drowsiness	0	1	2	3	4	5	6	7	8	9	10
26. Headaches	0	1	2	3	4	5	6	7	8	9	10
27. Feeling dizzy or lightheaded	0	1	2	3	4	5	6	7	8	9	10
28. Dry mouth	0	1	2	3	4	5	6	7	8	9	10
29. Altered sense of taste	0	1	2	3	4	5	6	7	8	9	10
30. Sore mouth or throat	0	1	2	3	4	5	6	7	8	9	10
31. Difficulty swallowing	0	1	2	3	4	5	6	7	8	9	10
32. Hair loss	0	1	2	3	4	5	6	7	8	9	10
33. Skin rash	0	1	2	3	4	5	6	7	8	9	10
34. Sore hands and/or feet	0	1	2	3	4	5	6	7	8	9	10
35. Numbness or pins and needles	0	1	2	3	4	5	6	7	8	9	10
36. Fevers	0	1	2	3	4	5	6	7	8	9	10
37. Hot flashes	0	1	2	3	4	5	6	7	8	9	10
38. Sweating or sweats	0	1	2	3	4	5	6	7	8	9	10
39. Problems with sex	0	1	2	3	4	5	6	7	8	9	10
40. Problems taking tablets	0	1	2	3	4	5	6	7	8	9	10
41. Problems with needles or injections	0	1	2	3	4	5	6	7	8	9	10
42. Inconvenience of treatment	0	1	2	3	4	5	6	7	8	9	10
43. Problems coping with treatment	0	1	2	3	4	5	6	7	8	9	10
44. Thought of actually having treatment	0	1	2	3	4	5	6	7	8	9	10
45. Problems looking after myself	0	1	2	3	4	5	6	7	8	9	10
46. Problems doing what I wanted	0	1	2	3	4	5	6	7	8	9	10
47. Problems for my friends or family	0	1	2	3	4	5	6	7	8	9	10

This is the end of the form. Thank you for filling it out.

NHMRC Clinical Trials Centre

TheraP: A randomised phase 2 trial of 177Lu-PSMA617 theranostic versus cabazitaxel in progressive metastatic castration resistant prostate cancer

Protocol number: ANZUP 1603

Statistical Analysis Plan

Prepared by: Andrew Martin
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Version: 2*

Version date: 28JAN20

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* Analgesic score was removed from the derivation of the pain response (Section 2.2) and pain progression (Section 2.6) endpoints in this version of the SAP because onsite monitoring revealed all sites to have had issues calculating analgesic score data correctly. The recommendation was made by the CTC Clinical Lead and supported by the Study Chair. The decision was not informed by any by-treatment analysis of the study data.

1 Introduction

The aim of the TheraP trial is to determine the activity and safety of Lu-PSMA versus cabazitaxel in men with metastatic prostate cancer progressing after multiple lines of chemotherapy and endocrine therapy.

The study is an open label phase II trial that randomises eligible patients to receive Lu-PSMA radionuclide therapy or cabazitaxel chemotherapy. Lu-PSMA-617 is given at 8.5 GBq IV every 6 weeks decreasing by 0.5 GBq for each cycle until: prohibitive toxicity, the patient is no longer thought to be benefiting, or post-therapy imaging with SPECT/CT shows complete resolution of PSMA uptake, to a maximum of 6 cycles. Treatment can be interrupted after an exceptional response, re-started at subsequent progression, and then continued to a maximum of 6 cycles in total. Cabazitaxel is given at 20mg/m2 IV every 3 weeks with prednisolone 10mg daily orally, continued until progressive disease or prohibitive toxicity to a maximum of 10 cycles

The randomisation is performed in a 1:1 ratio using minimisation with a random component stratified by: study site; volume of disease (>20 versus ≤20 sites of disease measured on ^{68}Ga-PSMA PET/CT); and, prior treatment with abiraterone or enzalutamide (yes vs no).

The target population is men with castration resistant prostate cancer suitable for chemotherapy with cabazitaxel, ECOG performance status 0-2, and a Ga-PSMA PET/CT scan showing high PSMA avidity without discordant disease on FDG PET/CT scan.

The primary objective is to determine the effect of Lu-PSMA on PSA response. The secondary objectives are to determine the effect of Lu-PSMA on: pain response, objective tumour response, progression-free survival, PSA progression-free survival, pain progression-free survival, radiographic progression-free survival, aspects of health related quality of life, overall survival, and adverse events. Correlative objectives include: to identify biomarkers that are prognostic and/or predictive of response to treatment, safety and resistance to study treatment.

2 Endpoint Derivation

A central clinical review will be performed on the results of the endpoint derivations specified below. A series of specific programmed endpoint checks are also performed, with any issues being centrally reviewed. Treatment allocation will not be included on the review outputs. Any instances where the specifications below are overruled on the basis of the clinical review findings will be documented.

Treatment with Lu-PSMA may be suspended if a participant exhibits an exceptional response to treatment, and re-treatment is permitted on disease progression (provided criteria in the protocol are met). In some cases this will mean a new baseline is applicable in the re-treatment period for some endpoints. For example, a participant that has Lu-PSMA suspended due to an exceptional response and is then re-treated with Lu-PSMA following radiological progression would have met the criteria for: (i) radiological progression as per the definition in 2.7 below (and be included in the analysis as such); and (ii) having a new baseline established for quantifying radiological progression-free survival in the re-treatment period. The new baseline would be the closest assessment up to,

and including, the date of the first dose of study treatment at re-treatment. The criteria for re-baselining are expected to be met infrequently, and any such endpoints will be summarised using simple descriptive statistics.

2.1 PSA Response (Primary Endpoint)

A PSA response is defined as a reduction of 50% or more PSA decline from baseline. The closest assessment up to, and including, the date of the first dose of study treatment will be used as the baseline reference value.

2.2 Pain Response

Pain is measured using the McGill-Melzack Present Pain Intensity scale (PPI). Pain Response is defined for patients with a baseline PPI score of ≥2 as a PPI score reduction of ≥2 points from baseline. The closest assessment up to, and including, the date of the first dose of study treatment will be used as the baseline reference value. Patients with a baseline PPI score of <2 will be excluded from the analyses of pain response.

2.3 Objective Tumour Response

Objective tumour response is defined as a confirmed complete response (CR) or partial response (PR) according to RECIST 1.1/PCWG3. The closest assessment up to, and including, the date randomisation will be used as the initial baseline reference value. If that is unavailable, the closest assessment up to, and including, the date of first dose of study treatment will be used as the initial baseline reference value.

2.4 Progression Free Survival

Progression free survival (PFS) is defined as the interval from the date of randomisation to the date of first evidence of PSA progression (See Section 2.5), pain progression (See Section 2.6), radiographic progression (See Section 2.7), or death from any cause, whichever occurs first, or the date of last known follow-up without progression (at which point the observation is censored). The censoring date will be the date of the latest PSA test.

2.5 PSA Progression Free Survival

PSA progression free survival (PSA PFS) is defined as the interval from the date of randomisation to the date of first evidence of PSA progression, commencement of non-protocol anti-cancer therapy, or death from any cause, whichever occurs first, or the date of last PSA test without PSA progression (at which point the observation is censored). PSA progression is defined as: a rise in PSA by at least 25% AND at least 2ng/mL above the nadir (lowest PSA point). This needs to be confirmed by a repeat PSA performed at least 3 weeks later. Early rises in PSA to 12 weeks will be disregarded in determining PSA progression if a new nadir is observed beyond 12 weeks. The closest assessment up to, and including, the date of the first dose of study treatment will be used as the initial PSA reference value.

2.6 Pain Progression-Free Survival

Pain progression-free survival (Pain PFS) is defined as the interval from the date of randomisation assessment to the date of first evidence of pain progression, commencement of non-protocol anti-cancer therapy, or death from any cause, whichever occurs first, or the date of last known follow-up without pain progression. Pain progression is defined as a ≥1 point increase in PPI score from nadir. Early rises in pain to 12 weeks will be disregarded in determining pain progression if a new nadir is observed beyond 12 weeks. The closest assessment up to, and including, the date of the first dose of study treatment will be used as the initial baseline reference value.

2.7 Radiographic Progression-Free Survival

Radiographic progression-free survival (rPFS) is defined as the interval from the date of randomisation to the date of first evidence of radiographic progression (PCWG3 criteria for bone lesions and RECIST 1.1 for soft tissue lesions), commencement of non-protocol anti-cancer therapy, or death from any cause, whichever occurs first, or the date of last known follow-up without radiographic progression. The closest assessment up to, and including, the date randomisation will be used as the initial baseline reference value. If that is unavailable, the closest assessment up to, and including, the date of first dose of study treatment will be used as the initial baseline reference value.

2.8 Overall Survival

Overall survival (OS) is defined as the interval from the date of randomisation to date of death from any cause, or the date of last known follow-up alive (at which point the observation is censored).

2.9 Quality of Life

The core EORTC QoL Questionnaire (QLQ-C30) will be scored according to standard conventions. Additional items on cancer specific symptoms will be collected using the patient Disease and Treatment Assessment (DATA) form. The closest assessment up to, and including, the date of the first dose of study treatment will be used as the initial baseline reference.

2.10 Safety

The NCI Common Terminology Criteria for Adverse Events Version 4.03 (NCI CTCAE) will be used to classify and grade the intensity of adverse events whilst on treatment and until 12 weeks after study treatment ends.

3 Analysis Sets

All randomised participants will be eligible for inclusion in the full analysis set in accordance with the intention-to-treat analysis principle. The full analysis set thus comprises the intention-to-treat (ITT) population. The ITT population will be the primary population used to evaluate the effect of Lu-PSMA on non-safety endpoints.

The Per-Protocol (PP) population may be used to evaluate the effect of Lu-PSMA on non-safety endpoints in sensitivity analyses. The PP population will comprise all randomised participants who receive at least one dose of assigned treatment (without subsequent treatment switching), and who are deemed eligible/evaluable on central clinical review (treatment allocation will not be included on outputs being reviewed)

Safety analyses will be performed using the safety population. The safety population will comprise all randomised participants who received at least one dose of study medication. If a patient receives at least one dose of Lu-PSMA (irrespective of randomised allocation) in the period between randomisation and cessation of study treatment, they will be included in the Lu-PSMA arm of the safety population. If that condition does not apply, and the patient receives a least one dose of cabazitaxel (irrespective of randomised allocation) in the period between randomisation and cessation of study treatment, they will be included in the cabazitaxel arm of the safety population. If neither of the above criteria apply, the patient will be excluded from the safety population.

The reasons for any exclusions from analysis sets will be documented.

4 Timing of Analyses

A series of three analyses is planned.

The first analysis will be performed based on a data cut-off date of: 31DEC19, or the date when all patients have accrued at least 3 months of follow-up to determine PSA response, whichever is greater. This will be the primary analysis for testing the null hypothesis on the primary endpoint of PSA response.

A second analysis will be performed based on a data cut-off date corresponding to when the last patient has completed protocol treatment, or 31DEC2020 is reached, whichever is sooner. This will be the primary analysis for testing the null hypothesis on the secondary endpoints of pain response, objective tumour response, and quality of life.

A third set of analyses will be undertaken on each of the PFS endpoints once 170 events have occurred, or 31DEC2020 is reached, whichever is sooner. This will be the primary analysis for testing the null hypothesis on the PFS endpoints (PFS, PSA PFS, Pain PFS, rPFS), and OS.

An analysis of the accumulating safety information will be performed on each occasion.

Analyses of endpoints may be undertaken early (prior to the timings of the primary analyses specified above), however the Haybittle-Peto rule will be used to interpret p-values from such analyses. This rule requires the test statistic to be more extreme than a benchmark of $\chi_1^2 = 9$. This benchmark is equivalent to a two-sided p-value of 0.0027.

5 Type I Error (Alpha)

A two-sided alpha of 5% will be applied to interpret the results of hypothesis tests and to construct confidence intervals. P-values from analyses of secondary endpoints that are unadjusted for multiple comparisons will be interpreted conservatively with the Benjamini-Hochberg procedure being used

as a guide where appropriate. See Section 4 on the use of the Haybittle-Peto rule to interpret the early analyses of endpoints.

6 Accounting for Stratification Factors

Randomisation is stratified using minimisation (with a random component) by: volume of disease (>20 versus ≤20 sites of disease measured on ^{68}Ga-PSMA PET/CT); prior treatment with abiraterone or enzalutamide (yes vs no); and, study site. Study sites will be grouped into three site categories ('NSW', 'Victoria', and 'Other'). If more than 90% of patients in an analysis set fall within a particular level of a stratification factor, that factor will not be used when performing stratified analyses (to minimise the likelihood of sparse stratification profiles). Sensitivity of conclusions given no accounting for stratification factors in the analysis will be explored.

7 Analysis of Study Endpoints and Patient Characteristics

In addition to the analyses described below being conducted for the analysis sets defined in Section 3, available data for screen failure patients (including reasons for ineligibility) will be reported descriptively.

7.1 Subject Disposition

The number of patients in the analysis sets (defined in Section 3) will be presented along with reasons for any exclusions. The Kaplan-Meier method will be used to summarise follow-up time for OS by treatment allocation with deaths being treated as censored observations. A CONSORT flow diagram will be prepared.

7.2 Baseline Demographic and Clinical Characteristics

Descriptive statistics will be prepared to summarise baseline characteristics (including PET imaging characteristics) of the study participants by treatment allocation.

7.3 Exposure to Study Medication

Reasons for discontinuations will be tabulated by treatment group.

7.4 Non-Protocol Anti-Cancer Treatment

The use of non-protocol anti-cancer treatment will be tabulated by treatment group. Available information on efficacy endpoints subsequent to initiation of non-protocol anti-cancer treatment will be summarised using descriptive statistics.

7.5 PSA Response (Primary Endpoint)

The primary analysis will be a comparison of the treatment groups on PSA response rate using a Cochran–Mantel–Haenszel $\chi 2$ test accounting for relevant stratification factors (see Section 6).

Multivariable logistic regression will be used to estimate odds ratios (with 95% CIs) for relevant stratification factors (see Section 6) and assigned treatment. A waterfall plot of PSA change will be prepared.

Patients with no assessment (e.g. due to consent withdrawal) beyond the first administration of study treatment will be considered as 'non-responders'. Exploratory analyses may however be used

to investigate the sensitivity of conclusions to multiple imputation techniques / tipping point analysis if more than 5% of the analysis set in a given treatment arm has no assessment beyond the first administration of study treatment.

7.6 Pain Response

Patient response will be analysed in the same way as PSA response.

7.7 Objective Tumour Response

Patient response will be analysed in the same way as PSA response.

7.8 Progression Free Survival (PFS)

PFS for each treatment group will be quantified using the Kaplan-Meier method and compared using a stratified log-rank test accounting for relevant stratification factors (see Section 6). Multivariable Cox PH regression will be used to estimate hazard ratios (with 95% CIs) for relevant stratification factors (see Section 6) and assigned treatment.

Patients with no post-randomisation information (e.g. due to consent withdrawal) will be censored in the analysis, however exploratory analyses may be used to investigate the sensitivity of conclusions to multiple imputation techniques / tipping point analysis if more than 5% of the analysis set in a given arm has no post-randomisation information.

7.9 PSA Progression Free Survival (PSA PFS)

PSA PFS will be analysed using the same approach as that described above for PFS.

7.10 Pain Progression-Free Survival (Pain PFS)

Pain Progression Free Survival will be analysed using the same approach as that described above for PFS.

7.11 Radiographic Progression-Free Survival (rPFS)

Radiographic Progression-Free Survival will be analysed using the same approach as that described above for PFS.

7.12 Overall Survival (OS)

OS will be analysed using the same approach as that described above for PFS.

7.13 Quality of life (QoL)

The QoL analyses will comprise the primary, secondary, and tertiary set of approaches next described.

7.13.1 Primary Approach: Deterioration-Free Survival (DetFS)

DetFS is defined as the time from randomisation to the first of the following events: a 10-point or more deterioration in health status from baseline (without subsequent 10-point or more

improvement compared with baseline, or an improvement to ≥90 if the baseline score was ≥90), progression, death, or treatment discontinuation. If a patient experiences none of these events, they will be censored. The censoring date will be the latest of the following: the date of the patient's last QoL assessment; and the date last known to be alive and (clinically) progression-free. Two DFS endpoints will be derived using different markers of health status deterioration based on the EORTC QLQ-C30: one used the Physical Function Scale ($DetFS_{PF}$), and the other used the General Health Scale ($DetFS_{GHS}$). The treatment groups will be compared on $DetFS_{PF}$ and DFS_{GHS} using a stratified log-rank test. Kaplan-Meier curves will be prepared and used to estimate median DetFS with 95% confidence intervals. An estimate of the hazard ratio for the treatment effect will be obtained using a Cox proportional hazard regression model with stratification factors included as covariates.

7.13.2 Secondary Approach: Mixed Model for Repeated Measures

Scale scores from the QLQ-C30 will be summarised by treatment group over time, and are planned to be analysed using a mixed model for repeated measures (MMRM). QoL data inadvertently collected >21 days after radiological progression will be excluded from the analysis. Time-point will be treated as a categorical covariate with QoL assessments being allocated to contiguous visit windows of 3-weeks duration. If two or more assessments for a given patient fall within the same window, that closest to the midpoint of the visit window will be retained. The MMRM will include covariates for baseline, treatment arm, post-baseline time-point, and a treatment-by-time point interaction. A blinded analysis (of interim) QoL data will be used to refine the analysis method. This will include: (i) specifying the covariance structure after evaluation of various options (including compound symmetry and autoregressive) using the AIC statistic; and (ii) specifying the strategy for accommodating any highly skewed data (e.g. log transformation, or split at median or other logical value with analysis of the resultant categorical endpoint performed using a repeated measures generalised linear model with a logit link function).

7.13.3 Tertiary Approach: Frequency of Troublesome Symptoms

The proportion of patients experiencing troublesome symptoms, or troublesome impacts on general aspects of QoL, as assessed by the patient Disease and Treatment Assessment (DATA) form will be calculated across the post-baseline assessment period (based on the worse grade reported during the on-treatment period), and comparisons between treatment arms will be made using logistic regression adjusting for baseline in exploratory analyses. Symptoms will be defined as troublesome if they are rated with an intensity of >= 3 points relative to 0="no trouble at all" and 10="worst I can imagine". Reductions on general aspects of QoL (e.g. Physical well-being) will be defined as troublesome if a 3-point decrement from the optimal score of 10 is reported (i.e. a rating of <=7 points relative to 0="worst possible" and 10="best possible").

7.14 Safety Data Analysis

Adverse Events (AEs) and Serious Adverse Events (SAEs) will be tabulated by treatment received (see Section 3) and CTCAE criteria including system organ class, term, and (worst) grade.

8 Analyses for Tertiary/Correlative Objectives

These analyses will be exploratory and the planned analyses methods are specified in only general terms herein.

The prognostic and/or predictive value of patient/disease characteristics will be explored using logistic regression for binary endpoints, and Cox Proportional Hazards regression for time-to-event endpoints. Tests of effect modification, to evaluate the predictive value of a characteristic, will be implemented by fitting the relevant characteristic-by-treatment interaction term to a given regression model (along with the associated main effect terms).

To simplify analyses, adjustment for stratification factors is not planned.

Comparisons between randomised groups and excluded participants (on basis of low PSMA avidity or discordant FDG-avid disease on screening PSMA/FDG PET) will be performed using linear modelling techniques taking into account imbalances in relevant prognostic factors.

Agreement between local and central FDG/PSMA eligibility determinations will be evaluated using the Kappa statistic.

9 Appendix

9.1 EORTC scoring

Instrument/Scale/Item					
	Scale	**No. items**	**Range**	**Item number**	**High score**
QLQ-C30					
Global health status/QoL	QL2	2	6	29, 30	+ve
Functional scales					
Physical	PF2	5	3	1 - 5	+ve
Role	RF2	2	3	6, 7	+ve
Emotional	EF	4	3	21 - 24	+ve
Cognitive	CF	2	3	20, 25	+ve
Social	SF	2	3	26, 27	+ve
Symptom scales / items					
Fatigue	FA	3	3	10, 12, 18	-ve
Nausea and vomiting	NV	2	3	14, 15	-ve
Pain	PA	2	3	9, 19	-ve
Dyspnoea	DY	1	3	8	-ve
Insomnia	SL	1	3	11	-ve
Appetite loss	AP	1	3	13	-ve
Constipation	CO	1	3	16	-ve
Diarrhoea	DI	1	3	17	-ve
Financial difficulties	FI	1	3	28	-ve

Definition:

In practical terms, if $I_1, I_2, \ldots, I_n$ are included in a scale, the procedure is as follows:

Calculate the raw score: $$RS = \frac{I_1 + I_2 + \cdots + I_n}{n}$$

Apply the linear transformation to 0-100 to obtain the score S:

Functional Scales: $$S = \{1 - \frac{(RS-1)}{range}\} * 100$$

Symptom scales/items: $$S = \{\frac{(RS-1)}{range}\} * 100$$

Global health status/QoL: $$S = \{\frac{(RS-1)}{range}\} * 100$$

Range is the difference between the maximum possible value of RS and the minimum possible value. The QLQ-C30 has been designed so that all the items in any scale take the same values. Therefore the range of RS equals the range of the items. Most items are scored 1 to 4 giving a *range* of 3. Exceptions are the items contributing to the global health status/QoL, which are 7-point questions with a *range* of 6.

Note a high score for a functional scale represents a *high/*healthy level of functioning, a high score for the global health status/QoL represents a *high QoL*, but a high score for a symptom/item scale represents a *high level of symptomatology/problems.*

Missing Items:

If at least half of the items from the scale have been answered, assume that the missing items have values equal to the average of those items which *are* present for that respondent.

Thus:

- Have at least half of the items from the scale been answered?
- If Yes, use all the items that were completed, and apply the standard equations for calculating the scale scores; ignore any items with missing values when making the calculations.
- If No, set scale score to missing.
- For single-item measures, set score to missing.